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Paper

Averaged Snapshots and Approximated Full Gradients for the Stochastic Variance-Reduced Gradient Method

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Abstract: Stochastic variance-reduced gradient (SVRG) converges linearly but requires full-gradient computations at each snapshot. No Full Grad SVRG (NFG-SVRG) avoids them at the cost of an approximation error of the full gradient that can cause an error floor. We propose Averaged Snapshots and Approximated full gradients for SVRG (ASA-SVRG), which uses the average of the inner-loop parameters as the snapshot, and show that this average minimizes the snapshot-dependent part of the upper bound on the approximation error. Experiments show a smaller approximation error than NFG-SVRG on a strongly convex problem, and SGD-level accuracy with less fluctuation on nonconvex image classification.

Key Words: Stochastic optimization, SVRG, iterate averaging, finite-sum minimization

1. Introduction

Many learning problems are formulated as the minimization of an empirical loss (objective function) over a large training set [1]. Let N be the number of data points and $\mathbf{w} \in \mathbb{R}^d$ the model parameters. The goal of learning is then to solve the following finite-sum minimization problem:

$$\min_{\mathbf{w}\in\mathbb{R}^d} f(\mathbf{w}) := \frac{1}{N}\sum_{n=1}^{N} f_n(\mathbf{w}). \tag{1}$$

Here, $f_n : \mathbb{R}^d \to \mathbb{R}$ denotes the loss on the n-th training sample and $f : \mathbb{R}^d \to \mathbb{R}$ denotes the empirical loss. The optimal solution is denoted by $\mathbf{w}^* = \arg\min_{\mathbf{w}} f(\mathbf{w})$.

A basic first-order method for Eq. (1) is gradient descent (GD) [2]. GD computes the gradient over all data at every update. It converges linearly when the objective is smooth and strongly convex, but its per-iteration cost grows with the number of data points N, which is prohibitive for large-scale data. To reduce this cost, stochastic gradient descent (SGD) [3, 4], which uses gradients computed from a subset of the data, is widely adopted. Each SGD iteration has a low computational cost, but the gradient computed from a subset generally retains variance. With a constant learning rate, the objective therefore stalls at a certain value instead of decreasing to the true optimum. This phenomenon is called the error floor. Since a decaying learning rate is required to obtain a high-accuracy solution, the convergence rate of SGD is generally sublinear even for smooth and strongly convex problems.

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Stochastic variance reduced gradient (SVRG) [5] explicitly reduces this variance. SVRG consists of an outer loop and an inner loop. At the beginning of each inner loop, it fixes a reference parameter, called the snapshot, and computes the gradient over all data at that point. This full gradient is used to correct the stochastic gradients, which reduces the variance and yields linear convergence with a constant learning rate for smooth and strongly convex problems. However, SVRG must compute the full gradient in every outer loop, and this cost can become a bottleneck for large-scale problems.

No Full Grad SVRG (NFG-SVRG) [6] was proposed to alleviate this problem. NFG-SVRG replaces the full gradient with the average of the stochastic gradients obtained in the previous outer loop (the average gradient), thereby avoiding a data-size-dependent computation in every outer loop. However, an approximation error remains between the average gradient and the true full gradient, so the objective generally converges to an error floor determined by this error. Hence, in schemes that approximate the full gradient, the choice of the snapshot affects the magnitude of the approximation error and, in turn, the upper bound on the attainable optimization accuracy.

In this paper, we clarify how the choice of the snapshot affects the upper bound on the approximation error, through the deviation between the snapshot and the inner-loop trajectory, in variance reduction methods that avoid full-gradient computation. We propose Averaged Snapshots and Approximated full gradients for the stochastic variance-reduced gradient method (ASA-SVRG), which uses the average of the parameters obtained in the inner loop (the average parameter) as the snapshot, and we discuss its convergence. Like NFG-SVRG, the proposed method avoids the full-gradient computation, and it stabilizes the reference point by parameter averaging. Since ASA-SVRG returns the average parameter, noise reduction similar to that of Polyak–Ruppert averaging can be expected. In addition, because the average is reset at every outer loop, the average taken late in training consists only of parameters near the optimum, which may lead to a better solution. The average parameter is the snapshot closest to the inner-loop trajectory, and it minimizes the deviation between the snapshot and the inner-loop trajectory, which is the snapshot-dependent part of the upper bound on the approximation error. In particular, we show that the average snapshot minimizes this snapshot-dependent part of the upper bound on the approximation error of ASA-SVRG, with the approximation error of SVRG being identically zero. We validate the proposed method on binary classification with the Mushroom dataset (a smooth and strongly convex setting) [7], image classification with AlexNet on CIFAR-10 [8], and image classification with ResNet18 on ImageWoof [15] (both nonconvex settings).

The contributions of this study are threefold:

- We propose ASA-SVRG, which uses the average parameter as the snapshot.
- We evaluate the convergence of the proposed and related methods under a unified workflow.
- We show that the approximation error of the full gradient is bounded by the sum of the variance of the stochastic gradients and the deviation between the snapshot and the inner-loop trajectory, and that the average parameter minimizes this deviation among the snapshots formed from the same inner-loop run.

In the experiments, the approximation error of ASA-SVRG is observed to be smaller than that of NFG-SVRG for binary classification on the Mushroom dataset. Moreover, in image classification with AlexNet on CIFAR-10 and with ResNet18 on ImageWoof, the approximation error of NFG-SVRG becomes so large that training breaks down, whereas ASA-SVRG avoids the full-gradient computation and keeps the approximation error at a practically negligible level. In particular, on ImageWoof, which violates the assumptions of the theoretical analysis, ASA-SVRG achieves accuracy comparable to that of SGD with a similar number of gradient evaluations, and its accuracy fluctuates less across iterations than that of SGD.

2. Preliminaries and related work

This section reviews the optimization methods on which the proposed method is based. We first outline gradient descent and stochastic gradient descent, then describe methods that reduce the variance of stochastic gradients, and finally introduce iterate averaging methods.

2.1 Gradient descent and stochastic gradient descent

The most basic optimization method for Eq. (1) is gradient descent (GD) [2]. At each iteration, GD computes the gradient over all N data points and updates the parameters as

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta \nabla f(\mathbf{w}_k). \tag{2}$$

Here, $\mathbf{w}_k$ denotes the parameters after k updates ($\mathbf{w}_0$ is random), η is the learning rate, and $\nabla f(\mathbf{w}_k) = \frac{1}{N}\sum_{n=1}^{N}\nabla f_n(\mathbf{w}_k)$ is the gradient computed over all data, called the full gradient; here, $\nabla f_n(\mathbf{w}_k) := \frac{\partial f_n(\mathbf{w}_k)}{\partial \mathbf{w}}$ denotes the gradient of the scalar-valued loss f_n with respect to the parameters $\mathbf{w}$, evaluated at $\mathbf{w}_k$. The full gradient reflects the information from the entire dataset and hence has no variance, but its computational cost depends on the number of data points N. When the objective is smooth and strongly convex, GD converges linearly with an appropriate constant learning rate [2]. However, computing the full gradient is expensive for large-scale data, which limits the practical scalability of GD.

To reduce the computational cost of GD, stochastic gradient descent (SGD) [3, 4] is widely used. SGD randomly draws a mini-batch $\mathcal{B}_k \subset \{1, \dots, N\}$ of data indices and updates the parameters as

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta \frac{1}{|\mathcal{B}_k|}\sum_{n\in\mathcal{B}_k}\nabla f_n(\mathbf{w}_k). \tag{3}$$

For simplicity, our theoretical analysis is restricted to a mini-batch size of one. The prior studies [5, 6] also develop their theory in the online setting. In the online setting, SGD draws a single index $n_k \in \{1, \dots, N\}$ and updates the parameters as

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta \nabla f_{n_k}(\mathbf{w}_k). \tag{4}$$

The gradient used in SGD is an approximation computed from a subset of the data, and is called the stochastic gradient. The stochastic gradient is an unbiased estimator of the full gradient, but its randomness due to data sampling gives rise to variance. Because of this variance, with a constant learning rate the objective does not decrease to the true optimum but stalls at a certain value. This phenomenon is called the error floor, and a decaying learning rate is required to obtain a high-accuracy solution. As a result, the convergence rate is generally sublinear even for smooth and strongly convex problems.

2.2 Stochastic variance reduced gradient methods

2.2.1 SVRG

Stochastic variance reduced gradient (SVRG) [5] explicitly reduces the variance of stochastic gradients. SVRG consists of an outer loop and an inner loop, and fixes a reference parameter at the beginning of each outer loop. This reference point is called the snapshot and serves as the reference for variance reduction. Algorithm 1 shows the SVRG algorithm. Let

Algorithm 1 SVRG

Require: Initial value $\mathbf{w}_0^0$, learning rate η, number of outer loops S, inner-loop length K

$\mathbf{z}_0 \leftarrow \mathbf{w}_0^0$
for $s = 0, 1, \dots, S-1$ **do**
  $\mathbf{w}_s^0 \leftarrow \mathbf{z}_s$
  $\mathbf{g}_s \leftarrow \nabla f(\mathbf{z}_s)$
  for $k = 0, 1, \dots, K-1$ **do**
    Draw index n_s^k
    $\mathbf{v}_s^k \leftarrow \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \mathbf{g}_s$
    $\mathbf{w}_s^{k+1} \leftarrow \mathbf{w}_s^k - \eta \mathbf{v}_s^k$
  end for
  Select $\mathbf{z}_{s+1}$ at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$
end for

Ensure: $\mathbf{z}_S$

s be the outer-loop index, k the inner-loop index, K the inner-loop length, and $\mathbf{z}_s$ the snapshot. The corrected gradient is defined as

$$\mathbf{v}_s^k = \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \mathbf{g}_s. \tag{5}$$

Here, n_s^k denotes the index of the data point drawn in outer loop s and inner loop k. The corrected gradient is a gradient estimator with reduced variance, obtained by subtracting the stochastic gradient at the snapshot from the stochastic gradient and adding the full gradient. In SVRG, $\mathbf{g}_s$ is the full gradient at the snapshot $\mathbf{z}_s$,

$$\mathbf{g}_s = \nabla f(\mathbf{z}_s) = \frac{1}{N}\sum_{n=1}^{N}\nabla f_n(\mathbf{z}_s), \tag{6}$$

and the parameters are updated by

$$\mathbf{w}_s^{k+1} = \mathbf{w}_s^k - \eta \mathbf{v}_s^k. \tag{7}$$

When the parameter $\mathbf{w}$ is close to the snapshot $\mathbf{z}_s$, the difference $\|\nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{z}_s)\|$ is small. Moreover, as $\mathbf{w}$ approaches the optimal solution $\mathbf{w}^*$, the variance of the corrected gradient decreases. For smooth and strongly convex problems, linear convergence with an appropriate constant learning rate has been shown [5].

However, SVRG requires the full gradient of Eq. (6) in every outer loop, and its cost depends on the number of data points N. For large-scale problems, this cost can become a bottleneck.

2.2.2 No Full Grad SVRG

No Full Grad SVRG (NFG-SVRG) [6] has been proposed as a variance reduction method that avoids the full-gradient computation. Algorithm 2 shows the NFG-SVRG algorithm. NFG-SVRG replaces $\mathbf{g}_s$ in Eq. (5), which is the full gradient in SVRG, with the

average of the stochastic gradients obtained in the previous outer loop,

$$\mathbf{g}_s = \frac{1}{K}\sum_{k=0}^{K-1} \nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k). \tag{8}$$

This average gradient substitutes for the full gradient and removes the need to compute the full gradient in every outer loop.

For SVRG-type methods, two ways of choosing the snapshot $\mathbf{z}_{s+1}$ of the next outer loop are known: selecting one uniformly at random from the inner-loop parameter sequence $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$, and adopting the final inner-loop parameter $\mathbf{w}_s^K$. Implementations most often adopt the final parameter. In contrast, our theoretical analysis is restricted to the former, that is, to selecting $\mathbf{z}_{s+1}$ at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$. This choice is justified for the following three reasons.

First, there is a technical constraint in the proof. In an analysis based on a telescoping sum of the inner-loop inequalities, what is naturally derived is the average error over all steps, $\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}[f(\mathbf{w}_s^k) - f(\mathbf{w}^*)]$. It is mathematically difficult to isolate and evaluate the error of the final point $\mathbf{w}_s^K$ alone.

Second, this evaluation serves as a conservative upper-bound guarantee and is standard in prior work. Random selection allows the average inner-loop objective gap to be expressed as the expected objective gap at the selected snapshot. In practice, the final inner-loop iterate is often used, although the present analysis does not establish the same bound for that iterate. Prior studies, including the SVRG paper [5], follow the same division of roles: the analysis assumes random selection, while the implementation uses the final parameter.

Third, the proposed ASA-SVRG directly closes this gap. ASA-SVRG uses the average of the inner-loop parameters as both the snapshot and the return value. By Jensen's inequality for convex functions, $f(\mathbf{z}_{s+1}) = f\left(\frac{1}{K}\sum_{k=0}^{K-1}\mathbf{w}_s^k\right) \le \frac{1}{K}\sum_{k=0}^{K-1} f(\mathbf{w}_s^k)$ holds. Therefore, an implementation that uses the average parameter is consistent with a theoretical analysis that evaluates the average error. For NFG-SVRG as well, we restrict the theoretical discussion to the case where $\mathbf{z}_{s+1}$ is selected at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$.

We define the difference between the full gradient and the average gradient as the approximation error. The approximation error quantifies how far the average gradient $\mathbf{g}_s$ deviates from the true full gradient $\nabla f(\mathbf{z}_s)$, and is given by

$$\mathbf{e}_s := \mathbf{g}_s - \nabla f(\mathbf{z}_s). \tag{9}$$

Algorithm 2 No Full Grad SVRG

Require: Initial value $\mathbf{w}_0^0$, learning rate η, number of outer loops S, inner-loop length K

$\mathbf{z}_0 \leftarrow \mathbf{w}_0^0$, $\mathbf{g}_0 \leftarrow \mathbf{0}$
for $s = 0, 1, \ldots, S-1$ **do**
  $\mathbf{w}_s^0 \leftarrow \mathbf{z}_s$
  $\bar{\mathbf{g}}_s^0 = \mathbf{0}$
  for $k = 0, 1, \ldots, K-1$ **do**
    Draw index n_s^k
    $\bar{\mathbf{g}}_s^{k+1} \leftarrow \frac{k}{k+1}\bar{\mathbf{g}}_s^k + \frac{1}{k+1}\nabla f_{n_s^k}(\mathbf{w}_s^k)$
    $\mathbf{v}_s^k \leftarrow \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \mathbf{g}_s$
    $\mathbf{w}_s^{k+1} \leftarrow \mathbf{w}_s^k - \eta\mathbf{v}_s^k$
  end for
  Select $\mathbf{z}_{s+1}$ at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$
  $\mathbf{g}_{s+1} = \bar{\mathbf{g}}_s^K$
end for

Ensure: $\mathbf{z}_S$

With this approximation error, the corrected gradient satisfies

$$\mathbb{E}_{n_s^k}\left[\mathbf{v}_s^k\right] = \nabla f(\mathbf{w}_s^k) + \mathbf{e}_s, \tag{10}$$

that is, the sum of the full gradient and the approximation error. SVRG uses the full gradient, so $\mathbf{e}_s = \mathbf{0}$ and no approximation error arises. In NFG-SVRG, in contrast, $\mathbf{e}_s \neq \mathbf{0}$ in general, and this approximation error persists. As a result, the objective value of NFG-SVRG converges to an error floor determined by the magnitude of the approximation error. The error floor is the phenomenon in which the objective cannot decrease below a certain value because of the approximation error; the smaller the approximation error, the lower the error floor.

2.3 Iterate averaging methods

Iterate averaging is another family of methods for suppressing the noise caused by stochastic gradients. Schedule-Free optimization [9], which removes the need for learning-rate scheduling, also uses averaging as the core of its stabilization.

2.3.1 Polyak–Ruppert averaging

Polyak–Ruppert averaging [10, 11] maintains the average of the parameter sequence independently of the parameter updates and returns this average parameter. The average parameter is the arithmetic mean of past parameters and provides a stable parameter estimate in which the influence of noise from stochastic gradients is smoothed out. In the general mini-batch setting, the update is given by

$$\begin{cases} \mathbf{w}_{k+1} = \mathbf{w}_k - \eta \dfrac{1}{|\mathcal{B}_k|} \displaystyle\sum_{n \in \mathcal{B}_k} \nabla f_n(\mathbf{w}_k), \\ \mathbf{x}_{k+1} = \dfrac{k}{k+1}\mathbf{x}_k + \dfrac{1}{k+1}\mathbf{w}_{k+1}. \end{cases} \tag{11}$$

Here, $\mathbf{x}_k$ is the average parameter up to step k, with $\mathbf{x}_0 = \mathbf{0}$. The second equation of Eq. (11) is a sequential update of the average parameter, which maintains the average without storing past parameters. Averaging the parameters smooths the noise due to stochastic gradients, and the parameter $\mathbf{x}_k$ used for inference becomes stable. However, the optimization trajectory itself proceeds on $\mathbf{w}_k$, so the learning process can still be affected by the noise of stochastic gradients.

2.3.2 Primal averaging

Primal averaging [12] also uses the average parameter at the point where the gradient is computed. In the general mini-batch setting, the update is given by

$$\begin{cases} \mathbf{w}_{k+1} = \mathbf{w}_k - \eta \dfrac{1}{|\mathcal{B}_k|} \displaystyle\sum_{n \in \mathcal{B}_k} \nabla f_n(\mathbf{x}_k), \\ \mathbf{x}_{k+1} = \dfrac{k}{k+1}\mathbf{x}_k + \dfrac{1}{k+1}\mathbf{w}_{k+1}. \end{cases} \tag{12}$$

This method stabilizes the learning trajectory itself by computing the gradient at the average parameter $\mathbf{x}_k$. On the other hand, it has been pointed out that random initialization can affect the average parameter and slow down convergence [9].

In summary, SVRG improves the accuracy of the estimator in every gradient update, whereas iterate averaging improves the accuracy of the solution finally obtained. The proposed method incorporates the insight of iterate averaging into the snapshot selection of NFG-SVRG, aiming to reduce the upper bound on the approximation error $\mathbf{e}_s$ of the full gradient and to improve the accuracy of the solution.

3. Proposed method

This section describes Averaged Snapshots and Approximated full gradients for the stochastic variance-reduced gradient method (ASA-SVRG)[1]. We then compare its convergence with that of the existing methods.

3.1 ASA-SVRG

SVRG can reduce the variance by means of the full gradient, but for large-scale data its computation becomes the bottleneck. NFG-SVRG replaces this computation with an average of stochastic gradients, but depending on how the snapshot is chosen, a large approximation error may remain and worsen the error floor to which the method converges. ASA-SVRG constructs the reference point for variance reduction as the average of the parameters obtained in the inner loop. Averaging smooths the fluctuations caused by mini-batch learning and keeps the deviation between the snapshot and the inner-loop trajectory small. The average parameter is the reference point closest to the inner-loop trajectory, and it minimizes this deviation, which is the snapshot-dependent part of the upper bound on the approximation error (Theorem 2). Therefore, the proposed method theoretically minimizes the snapshot-dependent part of the upper bound on the approximation error, which can lower the upper bound on the error floor, while avoiding the full-gradient computation.

ASA-SVRG uses the corrected gradient of Eq. (5) and, as in NFG-SVRG, approximates $\mathbf{g}_s$ by an average of stochastic gradients. The difference lies in the construction of the snapshot: ASA-SVRG takes the average of the parameters $\mathbf{w}_s^0, \ldots, \mathbf{w}_s^{K-1}$ obtained in the inner loop as the snapshot of the next outer loop. In the implementation, the average parameter and the average gradient are maintained by sequential updates, so there is no need to store all the inner-loop parameters. At inner-loop step k, we first update the running averages of the parameters and stochastic gradients as follows:

$$\bar{\mathbf{z}}_s^{k+1} = \frac{k}{k+1}\bar{\mathbf{z}}_s^k + \frac{1}{k+1}\mathbf{w}_s^k, \tag{13}$$

$$\bar{\mathbf{g}}_s^{k+1} = \frac{k}{k+1}\bar{\mathbf{g}}_s^k + \frac{1}{k+1}\nabla f_{n_s^k}(\mathbf{w}_s^k). \tag{14}$$

Then we compute the corrected gradient and update the parameters. At the end of each epoch, we set $\mathbf{z}_{s+1} = \bar{\mathbf{z}}_s^K$ and $\mathbf{g}_{s+1} = \bar{\mathbf{g}}_s^K$. Eqs. (13) and (14) yield the average of $\mathbf{w}_s^0, \ldots, \mathbf{w}_s^{K-1}$ and the average of the K stochastic gradients, respectively.

Algorithm 3 shows the ASA-SVRG algorithm. The initial average gradient is set to $\mathbf{g}_0 = \mathbf{0}$. In the first epoch, $\mathbf{w}_0^0 = \mathbf{z}_0$ and $\mathbf{g}_0 = \mathbf{0}$, so the corrected gradient is

$$\mathbf{v}_0^k = \nabla f_{n_0^k}(\mathbf{w}_0^k) - \nabla f_{n_0^k}(\mathbf{z}_0) + \mathbf{g}_0 = \mathbf{0} \tag{15}$$

and the parameters are not updated. Consequently, only the accumulation of the average gradient takes place in the first epoch. From the second epoch on, variance-reduced updates are performed using the accumulated average gradient, while the average parameter and the average gradient are updated sequentially.

[1] The implementation code and experimental results of the proposed method are available at https://github.com/yryo1005/ASAI_SVRG_programs.git.

Algorithm 3 ASA-SVRG

Require: Initial value $\mathbf{w}_0^0$, learning rate η, number of outer loops S, inner-loop length K

$\mathbf{z}_0 \leftarrow \mathbf{w}_0^0$, $\mathbf{g}_0 \leftarrow \mathbf{0}$
for $s = 0, 1, \ldots, S-1$ **do**
  $\mathbf{w}_s^0 \leftarrow \mathbf{z}_s$
  $\bar{\mathbf{z}}_s^0 \leftarrow \mathbf{0}$, $\bar{\mathbf{g}}_s^0 \leftarrow \mathbf{0}$
  for $k = 0, 1, \ldots, K-1$ **do**
    Draw index n_s^k uniformly
    $\bar{\mathbf{z}}_s^{k+1} \leftarrow \frac{k}{k+1}\bar{\mathbf{z}}_s^k + \frac{1}{k+1}\mathbf{w}_s^k$
    $\bar{\mathbf{g}}_s^{k+1} \leftarrow \frac{k}{k+1}\bar{\mathbf{g}}_s^k + \frac{1}{k+1}\nabla f_{n_s^k}(\mathbf{w}_s^k)$
    $\mathbf{v}_s^k \leftarrow \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \mathbf{g}_s$
    $\mathbf{w}_s^{k+1} \leftarrow \mathbf{w}_s^k - \eta \mathbf{v}_s^k$
  end for
  $\mathbf{z}_{s+1} \leftarrow \bar{\mathbf{z}}_s^K$, $\mathbf{g}_{s+1} \leftarrow \bar{\mathbf{g}}_s^K$
end for

Ensure: $\mathbf{z}_S$

The role of iterate averaging in ASA-SVRG becomes clear by contrast with two existing iterate averaging methods. Polyak–Ruppert averaging [10, 11] does not intervene in the update rule and reduces stochastic noise by using the average of the parameter sequence for inference. ASA-SVRG also returns the average parameter, so similar noise reduction can be expected, and it additionally uses this average as the snapshot of the next epoch. Because the average is reset at every outer loop, only the parameters near the optimum are averaged late in training, which may lead to a better solution. Primal averaging [12] computes the gradient at the average parameter and stabilizes the optimization trajectory itself. The stochastic gradient $\nabla f_{n_s^k}(\mathbf{z}_s)$ at the snapshot of ASA-SVRG plays a similar role to this practice of primal averaging, that is, computing the gradient at an average parameter, and the corrected gradient is evaluated along the averaged trajectory. Thus, ASA-SVRG incorporates iterate averaging into the SVRG framework, and the average parameter is a natural reference point for the average gradient. As discussed in Section 3.2, the average parameter minimizes the snapshot-dependent part of the upper bound on the approximation error of the full gradient.

3.2 Convergence characteristics and the effect of the snapshot

In this subsection, we give properties of the corrected gradient and the expected objective gap, and compare SVRG, NFG-SVRG, and ASA-SVRG. For comparison, we first show a unified workflow of SVRG-type methods in Algorithm 4. Each method sets $\mathbf{w}_s^0 = \mathbf{z}_s$ at the beginning of an outer loop and performs updates with the corrected gradient in the inner loop. The methods differ only in how $\mathbf{g}_s$ is computed and in the snapshot update rule. Note that the NFG-SVRG paper [6] uses the final inner-loop parameter as the snapshot, whereas our convergence analysis is restricted to the case where $\mathbf{z}_{s+1}$ is selected at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$. This treatment is justified by the analytical constraint of the telescoping-sum argument discussed in Section 2.2, and by the conservative upper-bound guarantee for the final parameter.

Algorithm 4 Unified workflow of SVRG-type methods

Require: Initial value $\mathbf{w}_0^0$, learning rate η, number of outer loops S, inner-loop length K

$\mathbf{z}_0 \leftarrow \mathbf{w}_0^0$
for $s = 0, 1, \ldots, S-1$ **do**
  $\mathbf{w}_s^0 \leftarrow \mathbf{z}_s$
  SVRG: $\mathbf{g}_s \leftarrow \nabla f(\mathbf{z}_s)$
  if $s = 0$ **then**
    NFG / ASA: $\mathbf{g}_0 \leftarrow \mathbf{0}$
  else
    NFG / ASA: $\mathbf{g}_s \leftarrow \frac{1}{K}\sum_{k=0}^{K-1} \nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k)$
  end if
  for $k = 0, 1, \ldots, K-1$ **do**
    Draw index n_s^k
    $\mathbf{v}_s^k \leftarrow \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \mathbf{g}_s$
    $\mathbf{w}_s^{k+1} \leftarrow \mathbf{w}_s^k - \eta \mathbf{v}_s^k$
  end for
  SVRG / NFG: select $\mathbf{z}_{s+1}$ at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$
  ASA: $\mathbf{z}_{s+1} \leftarrow \frac{1}{K}\sum_{k=0}^{K-1} \mathbf{w}_s^k$
end for

Ensure: $\mathbf{z}_S$

Table I shows the correspondence of the snapshot and the full-gradient estimate for SVRG, NFG-SVRG, and ASA-SVRG.

The following assumptions are made in the convergence analysis of this paper.

Assumption 1 (L-smoothness): Each f_n is L-smooth. That is, for any $\mathbf{w}, \mathbf{w}'$,

$$\|\nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{w}')\| \leq L\|\mathbf{w} - \mathbf{w}'\| \tag{16}$$

holds. Eq. (16) states that the rate of change of the gradient with respect to a change of the parameters is bounded by L. It follows that

Table I: Snapshot and full-gradient estimate of each method

Method	$\mathbf{z}_s$	$\mathbf{g}_s$
SVRG	Random selection from $\{\mathbf{w}_{s-1}^k\}_{k=0}^{K-1}$	$\nabla f(\mathbf{z}_s)$
NFG-SVRG	Random selection from $\{\mathbf{w}_{s-1}^k\}_{k=0}^{K-1}$	$\frac{1}{K}\sum_{k=0}^{K-1}\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k)$
ASA-SVRG	$\frac{1}{K}\sum_{k=0}^{K-1}\mathbf{w}_{s-1}^k$	$\frac{1}{K}\sum_{k=0}^{K-1}\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k)$

$$f_n(\mathbf{w}) - f_n(\mathbf{w}') - \langle \nabla f_n(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \le \frac{L}{2}\|\mathbf{w} - \mathbf{w}'\|^2 \tag{17}$$

holds. Eq. (17) is used in the subsequent Lemmas 1 and 2 to relate gradient differences to objective values.

Assumption 2 (convexity): Each f_n is convex. That is, for any $\mathbf{w}, \mathbf{w}'$ and any $\lambda \in [0,1]$, $f_n(\lambda\mathbf{w}+(1-\lambda)\mathbf{w}') \le \lambda f_n(\mathbf{w}) + (1-\lambda) f_n(\mathbf{w}')$ holds. Consequently, for any $\mathbf{w}, \mathbf{w}'$,

$$f_n(\mathbf{w}) \ge f_n(\mathbf{w}') + \langle \nabla f_n(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \tag{18}$$

holds. If each f_n is convex, the empirical loss f is also convex; this is used in Jensen's inequality for the average snapshot in ASA-SVRG and in the nonnegativity argument for the auxiliary function in Lemma 1. Moreover, since each f_n is convex and hence so is its average f, combining this with the L-smoothness of Assumption 1 gives, for any outer loop s,

$$\|\nabla f(\mathbf{z}_s)\|^2 \le 2L\big(f(\mathbf{z}_s) - f(\mathbf{w}^*)\big) \tag{19}$$

Eq. (19) is used in the proof of Lemma 2 to convert a gradient norm into an objective gap.

Assumption 3 (μ-strong convexity): f is μ-strongly convex. That is, for any $\mathbf{w}, \mathbf{w}'$,

$$f(\mathbf{w}) - f(\mathbf{w}') - \langle \nabla f(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \ge \frac{\mu}{2}\|\mathbf{w} - \mathbf{w}'\|^2 \tag{20}$$

holds. Eq. (20) states that the objective is sufficiently curved toward the optimum. In particular, setting $\mathbf{w}' = \mathbf{w}^*$ gives the following inequality:

$$f(\mathbf{w}) - f(\mathbf{w}^*) \ge \frac{\mu}{2}\|\mathbf{w} - \mathbf{w}^*\|^2 \tag{21}$$

Eq. (21) is used to convert the parameter distance into an objective gap.

The approximation error $\mathbf{e}_s$ of the full gradient is defined in Eq. (9). This quantity expresses how far the average gradient deviates from the true full gradient at the snapshot, and it governs the error floors of NFG-SVRG and ASA-SVRG. The corrected gradient can then be written as

$$\begin{aligned}\mathbf{v}_s^k &= \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \mathbf{g}_s \\ &= \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{z}_s) + \nabla f(\mathbf{z}_s) + \mathbf{e}_s\end{aligned} \tag{22}$$

where the second equality follows from Eq. (9). Taking the conditional expectation with respect to n_s^k further gives

$$\mathbb{E}_{n_s^k}\big[\mathbf{v}_s^k\big] = \mathbb{E}\big[\mathbf{v}_s^k \;\big|\; \mathbf{w}_s^k, \mathbf{z}_s, \mathbf{e}_s\big] = \nabla f(\mathbf{w}_s^k) + \mathbf{e}_s \tag{23}$$

In this paper, the subscripted expectation $\mathbb{E}_{n_s^k}[\cdot]$ denotes the conditional expectation with all quantities other than the data index n_s^k held fixed. That is, in Eq. (23) and in the subsequent discussion, the notation

$$\mathbb{E}_n\big[\cdot\big] = \mathbb{E}\big[\cdot \;\big|\; \text{fixed conditions}\big] \tag{24}$$

is used. In particular, Eq. (23) shows that the corrected gradient behaves as the full gradient plus the approximation error, and it is used in the expansion in Lemma 3. Note that Eq. (10) expresses the same relation as the right-hand side of Eq. (23).

Lemma 1 (Upper bound on the expected squared norm of gradient differences)**.** Under Assumptions 1 and 2, for any $\mathbf{w}$,

$$\mathbb{E}_n\big[\|\nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{w}^*)\|^2\big] \le 2L\big(f(\mathbf{w}) - f(\mathbf{w}^*)\big) \tag{25}$$

holds.

Eq. (25) states that the mean square of the differences between each sample gradient and the gradient at the optimum is bounded above by the objective gap. This lemma is used as a basic inequality when evaluating the second moment of the corrected gradient in Lemma 2. The proof is given in Appendix A.

Lemma 2 (Second moment of the corrected gradient)**.** Under Assumptions 1 and 2,

$$\begin{aligned}&\mathbb{E}_{n_s^k}\big[\|\mathbf{v}_s^k\|^2\big] \\ &\quad\le 6L\big(f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big) \\ &\qquad + 18L\big(f(\mathbf{z}_s) - f(\mathbf{w}^*)\big) + 6\|\mathbf{e}_s\|^2\end{aligned} \tag{26}$$

holds.

Eq. (26) states that the expected second moment of the corrected gradient is bounded by three terms: the objective gap at the current point, the objective gap at the snapshot, and the approximation error. This lemma is used in Lemma 3 to evaluate the distance after a parameter update. The proof is given in Appendix A.

The constants

$$c_1 = 2\eta(1-3\eta L), \qquad c_2 = 18\eta^2 L, \qquad c_3 = 6\eta^2 + \frac{\eta}{\mu} \tag{27}$$

are introduced. Here, c_1 is the coefficient contributing to the decrease of the objective gap, c_2 is the coefficient with which the snapshot gap remains as a residual, and c_3 is the coefficient with which the approximation error remains as a residual.

Lemma 3 (Upper bound on the cumulative expected objective gap)**.** Under $0 < \eta < 1/(3L)$,

$$\begin{aligned} c_1 \sum_{k=0}^{K-1} \mathbb{E}\big[f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big] \\ \le \left(Kc_2 + \frac{2}{\mu}\right) \mathbb{E}\big[f(\mathbf{z}_s) - f(\mathbf{w}^*)\big] \\ + Kc_3 \mathbb{E}\big[\|\mathbf{e}_s\|^2\big] \end{aligned} \tag{28}$$

holds.

Eq. (28) states that the sum of the objective gaps over the inner loop is bounded above by the snapshot gap and the approximation error. This lemma is the core used to derive the convergence recursion after the snapshot update. The proof is given in Appendix A.

Theorem 1 (Convergence characteristics of each method)**.** Under Assumptions 1–3, let $0 < \eta < 1/(3L)$, let the constants c_1, c_2, c_3 be given by Eq. (27), and let the contraction factor ρ be

$$\rho = \frac{Kc_2 + 2/\mu}{Kc_1} \tag{29}$$

Then

$$\mathbb{E}\big[f(\mathbf{z}_{s+1}) - f(\mathbf{w}^*)\big] \le \rho \mathbb{E}\big[f(\mathbf{z}_s) - f(\mathbf{w}^*)\big] + \frac{c_3}{c_1} \mathbb{E}\big[\|\mathbf{e}_s\|^2\big] \tag{30}$$

holds.

Eq. (30) states that the objective gap of the next epoch is bounded by the sum of a contraction term of the current epoch's gap and the approximation-error term.

Proof. When $\mathbf{z}_{s+1}$ is selected uniformly at random from $\{\mathbf{w}_s^k\}_{k=0}^{K-1}$,

$$\mathbb{E}\big[f(\mathbf{z}_{s+1}) - f(\mathbf{w}^*)\big] = \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}\big[f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big] \tag{31}$$

holds. Therefore

$$K\mathbb{E}\big[f(\mathbf{z}_{s+1}) - f(\mathbf{w}^*)\big] = \sum_{k=0}^{K-1} \mathbb{E}\big[f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big] \tag{32}$$

holds. For ASA, where $\mathbf{z}_{s+1} = \frac{1}{K}\sum_{k=0}^{K-1} \mathbf{w}_s^k$, f is convex by Assumption 2, so Jensen's inequality can be used:

$$f(\mathbf{z}_{s+1}) = f\left(\frac{1}{K} \sum_{k=0}^{K-1} \mathbf{w}_s^k\right) \le \frac{1}{K} \sum_{k=0}^{K-1} f(\mathbf{w}_s^k) \tag{33}$$

holds. Hence, similarly,

$$K\mathbb{E}\big[f(\mathbf{z}_{s+1}) - f(\mathbf{w}^*)\big] \le \sum_{k=0}^{K-1} \mathbb{E}\big[f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big] \tag{34}$$

holds. In either case, by Eq. (28),

$$\begin{aligned} Kc_1 \mathbb{E}\big[f(\mathbf{z}_{s+1}) - f(\mathbf{w}^*)\big] \\ \le \left(Kc_2 + \frac{2}{\mu}\right) \mathbb{E}\big[f(\mathbf{z}_s) - f(\mathbf{w}^*)\big] \\ + Kc_3 \mathbb{E}\big[\|\mathbf{e}_s\|^2\big] \end{aligned} \tag{35}$$

holds, and rearranging yields Eq. (30). □

Since $\mathbf{e}_s = \mathbf{0}$ in SVRG (Lemma 4), it converges linearly if η is chosen so that $\rho < 1$. In NFG-SVRG and ASA-SVRG, the second term on the right-hand side remains, and the objective converges to an error floor determined by the approximation error. Therefore, the relative merit of the three methods is determined by the second term on the right-hand side of Eq. (30), that is, by the magnitude of the approximation error $\mathbb{E}[\|\mathbf{e}_s\|^2]$. We bound this quantity in Theorem 2 below, and Theorem 3 shows that, among all snapshots formed from the same inner-loop run, the average parameter minimizes the snapshot-dependent part of this bound.

Lemma 4 (Approximation error of SVRG)**.** In SVRG,

$$\mathbf{e}_s^{\mathrm{SVRG}} = \mathbf{g}_s - \nabla f(\mathbf{z}_s) = \nabla f(\mathbf{z}_s) - \nabla f(\mathbf{z}_s) = \mathbf{0} \tag{36}$$

holds identically, so $\mathbb{E}[\|\mathbf{e}_s^{\mathrm{SVRG}}\|^2] = 0$.

Proof. In SVRG, the full gradient $\mathbf{g}_s = \nabla f(\mathbf{z}_s)$ itself is used as the snapshot gradient, so by Eq. (9), Eq. (36) holds identically, which gives $\mathbb{E}[\|\mathbf{e}_s^{\mathrm{SVRG}}\|^2] = 0$. □

Theorem 2 (Upper bound on the approximation error)**.** Under Assumption 1, for any inner-loop run $\{\mathbf{w}_{s-1}^k\}_{k=0}^{K-1}$ and any snapshot $\mathbf{z}_s$ formed from it (possibly with additional randomness, such as the uniform random selection used in NFG-SVRG),

$$\mathbb{E}[\|\mathbf{e}_s\|^2] \le \underbrace{\mathbb{E}\left[\frac{2}{K}\sum_{k=0}^{K-1}\|\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{w}_{s-1}^k)\|^2\right]}_{\text{term (i)}} + \underbrace{\mathbb{E}\left[\frac{2L^2}{K}\omega(\mathbf{z}_s)\right]}_{\text{term (ii)}} \tag{37}$$

holds, where

$$\omega(\mathbf{z}_s) := \sum_{k=0}^{K-1}\|\mathbf{w}_{s-1}^k - \mathbf{z}_s\|^2 \tag{38}$$

is the deviation between the snapshot and the inner-loop run. Here, term (i) is the variance of the stochastic gradients, which does not depend on the snapshot, and term (ii) is the deviation between the snapshot and the inner-loop run, scaled by $2L^2/K$, which depends on the snapshot.

Proof. $\mathbf{g}_s$ is the average of the stochastic gradients of the previous inner loop,

$$\mathbf{g}_s = \frac{1}{K}\sum_{k=0}^{K-1}\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) \tag{39}$$

Then the approximation error can be written as

$$\begin{aligned}\mathbf{e}_s &= \mathbf{g}_s - \nabla f(\mathbf{z}_s)\\ &= \frac{1}{K}\sum_{k=0}^{K-1}\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{z}_s)\\ &= \frac{1}{K}\sum_{k=0}^{K-1}\left(\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{w}_{s-1}^k)\right)\\ &\quad + \frac{1}{K}\sum_{k=0}^{K-1}\left(\nabla f(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{z}_s)\right)\end{aligned} \tag{40}$$

where we define

$$\begin{aligned}\mathbf{p}_{s-1}^k &:= \nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{w}_{s-1}^k),\\ \mathbf{q}_{s-1}^k &:= \nabla f(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{z}_s).\end{aligned} \tag{41}$$

Here, $\mathbf{p}_{s-1}^k$ is a term that does not depend on the snapshot $\mathbf{z}_s$, and $\mathbf{q}_{s-1}^k$ is a term that depends on the snapshot $\mathbf{z}_s$. By $\|\mathbf{a}+\mathbf{b}\|^2 \le 2\|\mathbf{a}\|^2 + 2\|\mathbf{b}\|^2$ and Jensen's inequality applied to the average over k,

$$\begin{aligned}\mathbb{E}\big[\|\mathbf{e}_s\|^2\big] &= \mathbb{E}\Big[\Big\|\frac{1}{K}\sum_{k=0}^{K-1}(\mathbf{p}_{s-1}^k + \mathbf{q}_{s-1}^k)\Big\|^2\Big]\\ &\le \frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\mathbf{p}_{s-1}^k + \mathbf{q}_{s-1}^k\|^2\big]\\ &\le \frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[2\|\mathbf{p}_{s-1}^k\|^2\big] + \frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[2\|\mathbf{q}_{s-1}^k\|^2\big]\\ &= \frac{2}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{w}_{s-1}^k)\|^2\big]\\ &\quad + \frac{2}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\nabla f(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{z}_s)\|^2\big]\end{aligned} \tag{42}$$

holds.

The first term in Eq. (42),

$$\frac{2}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\nabla f_{n_{s-1}^k}(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{w}_{s-1}^k)\|^2\big] \tag{43}$$

does not depend on the snapshot $\mathbf{z}_s$: whichever snapshot is chosen, this term remains unchanged for the same inner-loop run.

By Assumption 1, each f_n is L-smooth, and so is their average f. Hence the second term in Eq. (42) is bounded as

$$\begin{aligned}&\frac{2}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\nabla f(\mathbf{w}_{s-1}^k) - \nabla f(\mathbf{z}_s)\|^2\big]\\ &\le \frac{2L^2}{K}\sum_{k=0}^{K-1}\mathbb{E}\big[\|\mathbf{w}_{s-1}^k - \mathbf{z}_s\|^2\big]\end{aligned} \tag{44}$$

that is, writing $\omega(\mathbf{z}_s)$ for the sum on the right-hand side as in the statement, the second term in Eq. (42) $\le \frac{2L^2}{K}\omega(\mathbf{z}_s)$. Combining this with Eq. (42) gives Eq. (37), as claimed. □

Term (i) is common to all methods that use the average gradient. The comparison below therefore concerns only term (ii), the controllable, snapshot-dependent part of the upper bound, and we make no claim about the relative magnitude of terms (i) and (ii).

Theorem 3 (Minimization of the deviation between the snapshot and the trajectory by the average snapshot)**.** For any inner-loop run $\{\mathbf{w}_{s-1}^k\}_{k=0}^{K-1}$, let $\mathbf{z}_s^{\mathrm{ASA}} := \frac{1}{K}\sum_{k=0}^{K-1}\mathbf{w}_{s-1}^k$, and let $\mathbf{z}_s$ be any snapshot formed from that run, possibly with additional randomness, such as the uniform random selection used in NFG-SVRG, which selects $\mathbf{z}_s^{\mathrm{NFG}}$ uniformly at random from $\{\mathbf{w}_{s-1}^k\}_{k=0}^{K-1}$ as a special case of $\mathbf{z}_s$. Then

$$0 \le \mathbb{E}\Big[\frac{2L^2}{K}\omega(\mathbf{z}_s^{\mathrm{ASA}})\Big] \le \mathbb{E}\Big[\frac{2L^2}{K}\omega(\mathbf{z}_s)\Big] \tag{45}$$

holds.

Therefore, the average parameter minimizes, with respect to the snapshot, the right-hand side of Eq. (37) in Theorem 2, that is, the upper bound on $\mathbb{E}[\|\mathbf{e}_s\|^2]$. Since the error floor in Eq. (30) is governed by $\mathbb{E}[\|\mathbf{e}_s\|^2]$, the average snapshot minimizes the snapshot-dependent part of the upper bound on the error floor.

Theorem 3 concerns only the snapshot-dependent part of the upper bound for a given inner-loop run; it does not assert an ordering between the actual approximation errors of independently trained models, which is examined empirically in Section 4. The snapshot is also the initial parameter of the next inner loop and therefore affects its trajectory; the present analysis compares methods within a single inner loop, and the effect of the choice of snapshot on the stochastic-gradient variance and the approximation error over multiple outer loops is instead analyzed experimentally in Section 4, with a theoretical analysis left for future work.

Proof. Expanding $\omega(\mathbf{z}_s)$ gives

$$\begin{aligned}\omega(\mathbf{z}_s) &= \sum_{k=0}^{K-1}\big(\|\mathbf{w}_{s-1}^k\|^2 - 2\langle\mathbf{w}_{s-1}^k, \mathbf{z}_s\rangle + \|\mathbf{z}_s\|^2\big)\\ &= \sum_{k=0}^{K-1}\|\mathbf{w}_{s-1}^k\|^2 - 2\Big\langle\mathbf{z}_s, \sum_{k=0}^{K-1}\mathbf{w}_{s-1}^k\Big\rangle + K\|\mathbf{z}_s\|^2\end{aligned} \tag{46}$$

Setting $\dfrac{\mathrm{d}\omega(\mathbf{z}_s)}{\mathrm{d}\mathbf{z}_s} = \mathbf{0}$ yields

$$-2\sum_{k=0}^{K-1}\mathbf{w}_{s-1}^k + 2K\mathbf{z}_s = \mathbf{0} \quad\Rightarrow\quad \mathbf{z}_s^* = \frac{1}{K}\sum_{k=0}^{K-1}\mathbf{w}_{s-1}^k \tag{47}$$

This minimizer coincides with the average parameter, that is, $\mathbf{z}_s^* = \mathbf{z}_s^{\mathrm{ASA}}$. Since ω is a strictly convex quadratic function of $\mathbf{z}_s$, this stationary point is its unique global minimizer, so $\omega(\mathbf{z}_s^{\mathrm{ASA}}) \le \omega(\mathbf{z}_s)$ for any $\mathbf{z}_s$; this inequality holds for every realization of the inner-loop run $\{\mathbf{w}_{s-1}^k\}_{k=0}^{K-1}$ and of the snapshot $\mathbf{z}_s$, however the latter is determined. Taking the expectation, and noting that $\omega(\cdot) \ge 0$, gives Eq. (45). □

In summary, the following holds.

- Since $\mathbf{e}_s = \mathbf{0}$ (Lemma 4), SVRG converges linearly with $\rho < 1$ under an appropriate η.
- Because of the approximation error, the objective of NFG-SVRG and ASA-SVRG converges to an error floor.
- By Theorem 2, the upper bound on $\mathbb{E}[\|\mathbf{e}_s\|^2]$ is expressed as the sum of the variance of the stochastic gradients (term (i)) and the deviation between the snapshot and the inner-loop run (term (ii)).
- By Theorem 3, the average snapshot minimizes this deviation, that is, the snapshot-dependent part of the upper bound on the error term in Eq. (30).

4. Experiments

To verify the validity of the theoretical analysis and the practical effectiveness of the proposed method, we conduct three experiments with different characteristics. The first experiment deals with a smooth and strongly convex problem that strictly satisfies Assumptions 1–3 of the theoretical analysis, and checks whether the reduction of the deviation between the snapshot and the inner-loop trajectory achieved by ASA-SVRG (Theorem 3) is also reflected in the actual approximation errors. The second and third experiments deal with nonconvex neural network training problems that do not satisfy the assumptions of the theory, and evaluate the practical effectiveness of the proposed method. In addition, an experiment on a small-scale problem that is separate from the theoretical analysis and reproduces the supplementary experiment of the NFG-SVRG paper [6], and additional experiments with different mini-batch sizes for the second and third experiments are presented in Appendix B. In each of the following figures, the solid lines show the mean over random seeds and the shaded regions show the standard error of the mean.

In all experiments, the horizontal axis is #grad/N, the number of gradient computations normalized by the number of training data points. The number of gradient computations required per epoch differs among the methods; in terms of #grad/N, it is 1 for SGD,

2 for NFG-SVRG and ASA-SVRG, and 3 for SVRG. Therefore, the number of outer loops (epochs) of each method was set individually so that #grad$/N$ reaches the target value defined for each experiment. In all experiments, we perform trials with five random seeds and evaluate the results by the mean and the standard error. The random seed determines the initial parameters of the model and the order in which the data are drawn; for the Mushroom and a9a datasets, it also determines how the data are split into training and validation sets. The approximation error of the full gradient is computed as the squared norm of the difference between $\mathbf{g}_s$ and the true full gradient $\nabla f(\mathbf{z}_s)$ at the start of each epoch. Although NFG-SVRG and ASA-SVRG do not otherwise require the full gradient, it is computed solely to evaluate this approximation error, and this additional computation is not included in the count of gradient computations.

4.1 Binary classification on the Mushroom dataset

As the first experiment, we compare the methods on a smooth and strongly convex problem that strictly satisfies Assumptions 1–3 of the theoretical analysis. This experiment verifies the proposed method under the same problem setting as the convergence analysis in Section 3.

In this section and in the BCE-loss experiment of Appendix B.1, we consistently use $\eta = 0.99/(3L)$, close to the upper bound of the condition $0 < \eta < 1/(3L)$ of Lemma 3, and $\rho < 1$ does not necessarily hold for this η (indeed, $\rho > 1$ in these experiments). Consequently, the rapid decrease of the objective observed in these experiments is not itself the linear convergence (the contraction under $\rho < 1$) guaranteed by Theorem 1. However, the bound ρ in Eq. (30) is obtained using inequalities such as the Cauchy–Schwarz and Young inequalities used in the proofs of Lemmas 2 and 3, and it does not provide a lower bound on the actual convergence rate. That is, $\rho \geq 1$ only means that the upper bound obtained in the proof of Theorem 1 diverges; it does not contradict the algorithm converging in practice, nor the objective decreasing in a way that looks like linear convergence.

We use the Mushroom dataset of the UCI Machine Learning Repository [7]. The task is to classify mushrooms as edible or poisonous from 22 discrete features such as shape, color, and odor. The dataset has 8124 samples in total, and ordinal encoding, which converts each categorical feature to an integer, keeps the feature dimension at $D = 22$. All data are split 9:1 into training and validation sets, and are standardized with the statistics of the training data before use. The numbers of training and validation samples are $N_{\text{train}} = 7311$ and $N_{\text{test}} = 813$, respectively.

The model is a linear combination model with an intercept b, and binary classification is performed by logistic regression. The objective function is the binary cross-entropy loss with L2 regularization on both the weight $\mathbf{w}$ and the intercept b, defined as

$$f(\mathbf{w}, b) = \frac{1}{N_{\text{train}}} \sum_{n=1}^{N_{\text{train}}} \ell_{\text{BCE}}(y_n, \hat{y}_n(\mathbf{w}, b)) + \frac{\lambda}{2}\left(\|\mathbf{w}\|^2 + b^2\right) \tag{48}$$

where $\ell_{\text{BCE}}(y, \hat{y}) = -[y \log \hat{y} + (1-y)\log(1-\hat{y})]$ is the binary cross-entropy loss and $\hat{y}_n(\mathbf{w}, b) = \sigma(\mathbf{w}^\top \mathbf{x}_n + b)$ is the prediction using the sigmoid function σ. Regularizing the intercept b as well makes μ-strong convexity with $\mu = \lambda$ hold uniformly for all parameters $(\mathbf{w}, b)$, so that Assumption 3 is strictly satisfied. The smoothness constant is computed from an upper bound on the Hessian of the loss of a single sample as $L = \frac{1}{4}\max_n \|\tilde{\mathbf{x}}_n\|^2 + \lambda$, where $\tilde{\mathbf{x}}_n$ is the feature vector augmented with a constant 1 corresponding to the intercept. The regularization coefficient λ is set as 3×10^{-2} times the smoothness constant without regularization, $L_0 = \frac{1}{4}\max_n \|\tilde{\mathbf{x}}_n\|^2$, which gives $L = 31.4355$ and $\lambda = \mu = 0.9156$.

We compare four methods: SGD, SVRG, NFG-SVRG, and the proposed ASA-SVRG. SVRG and NFG-SVRG adopt the final inner-loop parameter as the snapshot, and ASA-SVRG adopts the average parameter. The inner loop is online learning with mini-batch size one that passes over the training data once per epoch by random reshuffling, and the inner-loop length is $K = N_{\text{train}} = 7311$. The number of outer loops S of each method was set so that #grad$/N$ is approximately 30. The learning rate is $\eta = 0.99/(3L) = 1.0498 \times 10^{-2}$, which is close to the upper bound of the condition $0 < \eta < 1/(3L)$ of Lemma 3. The optimal value $f(\mathbf{w}^*)$ is estimated with high accuracy by L-BFGS-B.

Figure 1(A) shows the objective error $f(\mathbf{z}_s) - f(\mathbf{w}^*)$. SGD converges to an error floor under a constant learning rate and stalls at an almost constant level after the early stage of training. In contrast, SVRG, NFG-SVRG, and ASA-SVRG all exhibit linear convergence, and the objective error reaches almost 0. However, the convergence speed differs among the methods, and ASA-SVRG converges faster than SVRG and NFG-SVRG.

Figure 1(B) shows the classification accuracy on the validation data. The accuracy curve of NFG-SVRG

coincides exactly with that of ASA-SVRG and appears overlapped in the plot. Since the corrected gradient is 0 in the first outer loop of these methods, the accuracy stays near the level of random guessing until #grad/N reaches 2. At the end of training, the accuracies of SVRG, NFG-SVRG, and ASA-SVRG coincide and are slightly lower than that of SGD.

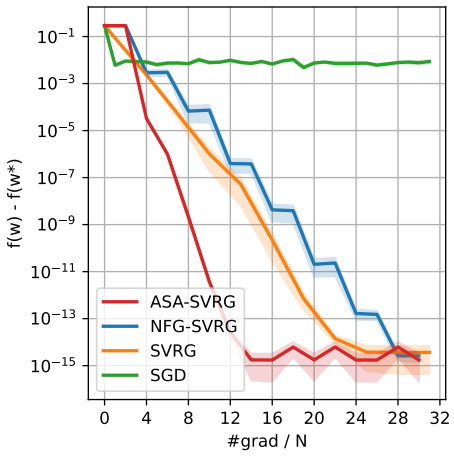
Figure 1(C) shows the approximation error $\|\mathbf{e}_s\|^2$ of the full gradient for ASA-SVRG and NFG-SVRG. In the early stage of training, the approximation error of ASA-SVRG is consistently lower than that of NFG-SVRG, with a difference of several orders of magnitude. As training proceeds, the approximation error of ASA-SVRG decreases to the level of rounding error and approaches that of SVRG (identically 0). The observed approximation error of ASA-SVRG being smaller than that of NFG-SVRG is consistent with the effect of reducing the deviation between the snapshot and the inner-loop trajectory (Theorem 3), in a problem setting that satisfies the assumptions of the theoretical analysis. The approximation error of SVRG remains identically zero, consistent with Lemma 4.

4.2 Multiclass classification on CIFAR-10 with AlexNet

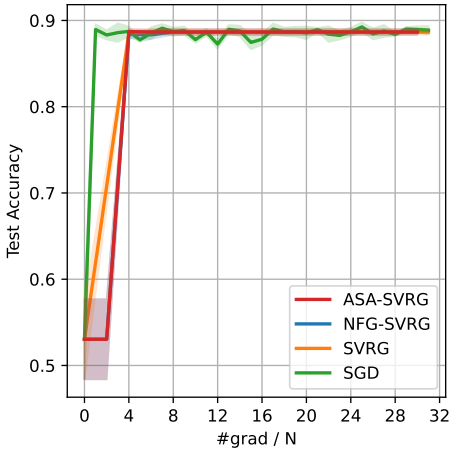
As the second experiment, we compare the methods on a nonconvex problem that does not satisfy Assumptions 1–3 of the theory, and evaluate the practical effectiveness of the proposed method. Unlike the theoretical setting targeted by the convergence analysis in Section 3, this experiment is a case that assumes a practical application using a deep neural network.

We use CIFAR-10 [8]. CIFAR-10 is a 10-class image classification dataset consisting of 32×32 color images, with $N_{\text{train}} = 50000$ training samples and $N_{\text{test}} = 10000$ validation samples. Each channel is normalized using the means $(0.4914, 0.4822, 0.4465)$ and the standard deviations $(0.2470, 0.2435, 0.2616)$ of the training data. No data augmentation is used.

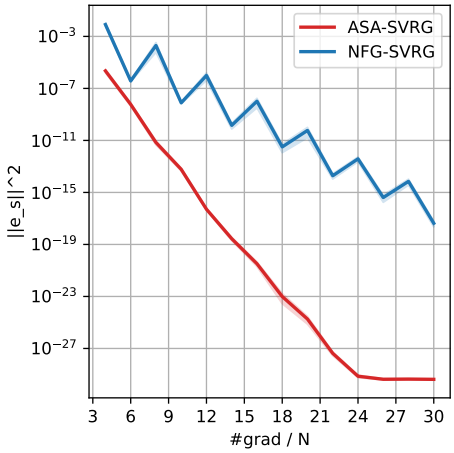
The model is a convolutional neural network based on AlexNet [13]. The network consists of five convolutional layers (with 64, 192, 384, 256, and 256 output channels, in this order) and three fully connected layers (with output dimensions 1024, 512, and 10, in this order). The first and second convolutional layers have kernel size 5 and padding 2, and the third to fifth have kernel size 3 and padding 1; each convolutional layer is immediately followed by layer normalization [14] and a ReLU activation, and max pooling with window size 2 is applied after the first, second, and fifth layers. The layer normalization used here normalizes along the channel dimension at each spatial location



(A)



(B)



(C)

Fig. 1: Results on the Mushroom dataset: (A) objective error $f(\mathbf{z}_s) - f(\mathbf{w}^*)$, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

and does not depend on other samples in the batch. We do not use batch normalization, dropout, or data aug-

mentation in this experiment. These change the output for identical parameters depending on the mini-batch composition and the timing of inference, so the values of the full gradient and the average gradient fluctuate depending on the mini-batch drawn, and the approximation error may grow for reasons unrelated to the performance of the optimizer. Therefore, we adopt layer normalization, which is a deterministic normalization whose statistics are determined per sample. The objective function is the multiclass cross-entropy loss with L2 regularization on the weights of the convolutional and fully connected layers, defined as

$$f(\mathbf{w}) = \frac{1}{N}\sum_{n=1}^{N} \ell_{\mathrm{CCE}}(\mathbf{y}_n, \hat{\mathbf{y}}_n(\mathbf{w})) + \frac{\lambda}{2}\sum_{l} \|\mathbf{W}_l\|^2 \quad (49)$$

where $\ell_{\mathrm{CCE}}(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{c=1}^{10} y_c \log \hat{y}_c$ is the multiclass cross-entropy loss, $\mathbf{W}_l$ is the weight of the l-th convolutional or fully connected layer, and the regularization coefficient is $\lambda = 5 \times 10^{-4}$. Owing to the nonlinearity of the neural network, the objective function is nonconvex and does not satisfy Assumptions 1–3 of the theoretical analysis.

We compare four methods: SGD, SVRG, NFG-SVRG, and ASA-SVRG. The mini-batch size is 128, and the inner-loop length is the number of mini-batches per epoch, $K = \lceil N_{\mathrm{train}}/128 \rceil = 391$. The learning rate is $\eta = 1 \times 10^{-3}$. The number of outer loops S of each method was set so that $\#\mathrm{grad}/N$ is approximately 90.

Figure 2(A) shows the training loss. The training loss decreases fastest for SGD, followed by ASA-SVRG, while that of SVRG decreases more slowly, and it barely decreases for NFG-SVRG after the early stage of training. Moreover, the training loss of ASA-SVRG fluctuates less from iteration to iteration than that of SGD and decreases smoothly.

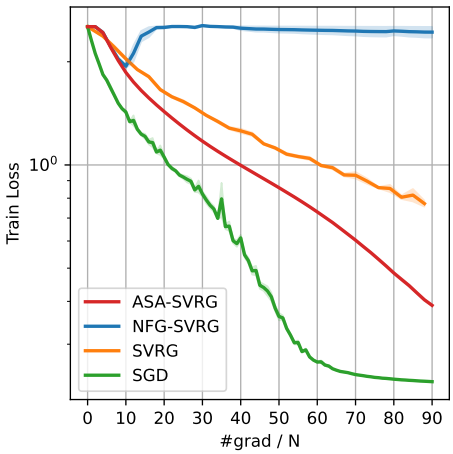
Figure 2(B) shows the classification accuracy on the validation data. With $\#\mathrm{grad}/N$ on the horizontal axis, the accuracy is highest for SGD, followed by ASA-SVRG and then SVRG. NFG-SVRG is far below the other three methods and also shows markedly larger variability across trials. ASA-SVRG avoids the full-gradient computation that SVRG requires in every outer loop, yet reaches higher accuracy than SVRG at the same $\#\mathrm{grad}/N$. As in the training loss of (A), the curve of ASA-SVRG fluctuates less from iteration to iteration than that of SGD, and the accuracy improves more stably. Unlike SVRG, ASA-SVRG cannot make the approximation error exactly zero, so the result that it reaches higher accuracy than SVRG at the same $\#\mathrm{grad}/N$ may seem counterintuitive. However, ASA-SVRG returns the average of the inner-loop parameters, which, like Polyak–Ruppert averaging, smooths the noise of the stochastic gradients and improves the accuracy of the returned solution. Therefore, in ASA-SVRG, the small bias caused by approximating the full gradient appears to be compensated by the noise reduction due to parameter averaging, so that it achieves a training curve better than that of SVRG without the cost of the full-gradient computation that SVRG incurs in every outer loop.

Figure 2(C) shows the approximation error $\|\mathbf{e}_s\|^2$. The approximation error of NFG-SVRG fluctuates over a wide range across epochs, and this gap from epoch to epoch is large enough that the training of NFG-SVRG effectively breaks down; for most of training, the approximation error of ASA-SVRG remains far below that of NFG-SVRG. In some intervals in the middle of training, the approximation error of NFG-SVRG momentarily falls below that of ASA-SVRG. However, as shown in (B), this results from the accuracy of NFG-SVRG having already dropped to a level close to random guessing at that point and its training being almost stagnant, and does not mean that NFG-SVRG provides a good approximation. A tendency qualitatively consistent with the effect of reducing the deviation between the snapshot and the inner-loop trajectory (Theorem 3), that the approximation error of ASA-SVRG is far below that of NFG-SVRG, is thus observed even in this nonconvex setting. In NFG-SVRG, the accumulated approximation error appears to affect the whole training as a bias of the corrected gradient, manifesting itself as a drop in accuracy and instability across trials. In contrast, ASA-SVRG keeps the same computational efficiency as NFG-SVRG by avoiding the full-gradient computation, suppresses the approximation error to a practically negligible level by parameter averaging, and achieves training performance equal to or better than that of SVRG, which uses the full gradient.

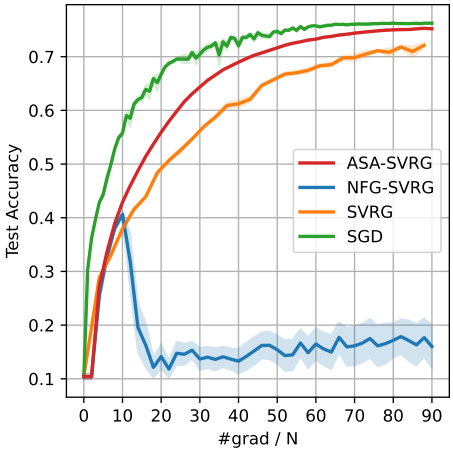
In additional experiments with mini-batch sizes 32 and 512, a tendency similar to the above was observed for mini-batch size 512, whereas for mini-batch size 32 the training of both NFG-SVRG and ASA-SVRG broke down as the inner-loop length K increased. Details are given in Appendix B.2.

4.3 Multiclass classification on ImageWoof with ResNet18

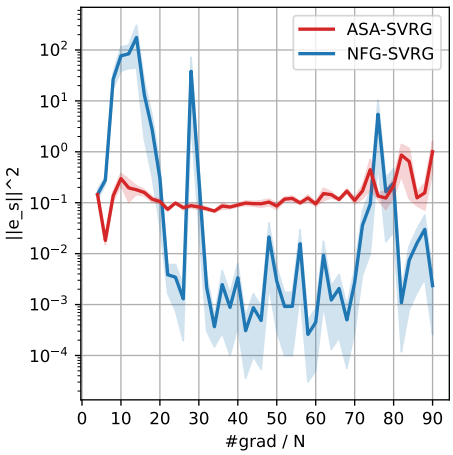
As the third experiment, we compare the methods on a larger and more difficult nonconvex problem than the



(A)



(B)



(C)

Fig. 2: Results on CIFAR-10 (mini-batch size 128): (A) training loss, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

second experiment, and further verify the practical effectiveness of the proposed method. This experiment is also a case that assumes a practical application using a deep neural network, unlike the theoretical setting targeted by the convergence analysis in Section 3.

We use ImageWoof [15]. ImageWoof is an image classification dataset of 10 dog-breed classes extracted from ImageNet, with $N_{\text{train}} = 9025$ training samples and $N_{\text{test}} = 3929$ validation samples. Each image is resized so that its shorter side is 256 pixels, center-cropped to 224×224 pixels, and normalized by the channel-wise mean and standard deviation computed from the training data. No data augmentation such as random cropping or horizontal flipping is used.

The model is ResNet18 [16]. The network architecture is the standard ResNet18 (without changing the residual block structure, the number of channels, or the skip connections), and only the output layer is replaced for 10-class classification. In addition, all batch normalization layers in the model are replaced with layer normalization [14] layers, which do not depend on other samples in the batch. The reason is the same as in Section 4.2: to prevent normalization that depends on batch statistics from making the full gradient fluctuate with the mini-batch composition and thereby increasing the approximation error for reasons unrelated to the performance of the optimizer. The objective function is the multiclass cross-entropy loss with L2 regularization on the weights of the convolutional and fully connected layers, the same as Eq. (49). The regularization coefficient is $\lambda = 5 \times 10^{-4}$.

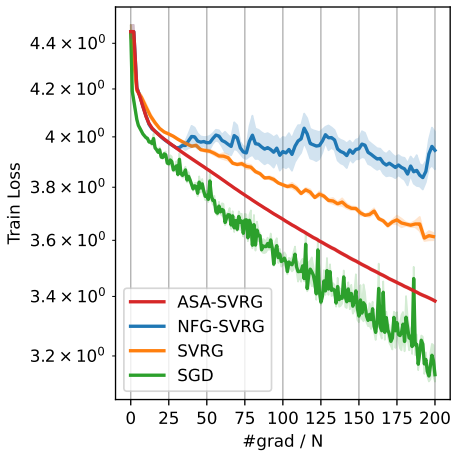
We compare four methods: SGD, SVRG, NFG-SVRG, and ASA-SVRG. The mini-batch size is 128, and the inner-loop length is the number of mini-batches per epoch, $K = \lceil N_{\text{train}}/128 \rceil = 71$. The learning rate is $\eta = 1 \times 10^{-3}$. The number of outer loops S of each method was set so that $\#\text{grad}/N$ is approximately 200.

Figure 3(A)–(C) show the training loss, the classification accuracy on the validation data, and the approximation error $\|\mathbf{e}_s\|^2$. At $\#\text{grad}/N = 200$, the accuracies of SGD and ASA-SVRG reach almost the same level, followed by SVRG, and NFG-SVRG is far below the other three methods. As in the training loss of (A), the accuracy curve of ASA-SVRG fluctuates less from iteration to iteration than that of SGD, and improves more stably. At the same point, the approximation error of ASA-SVRG is far below that of NFG-SVRG. In this experiment, a consistent advantage over NFG-SVRG is observed in both approximation error and classification accuracy, indicating that ASA-SVRG trains more stably than NFG-SVRG, which also avoids the full-gradient computation. In additional experiments with mini-batch sizes 512 and 32, a tendency similar to the above was observed for

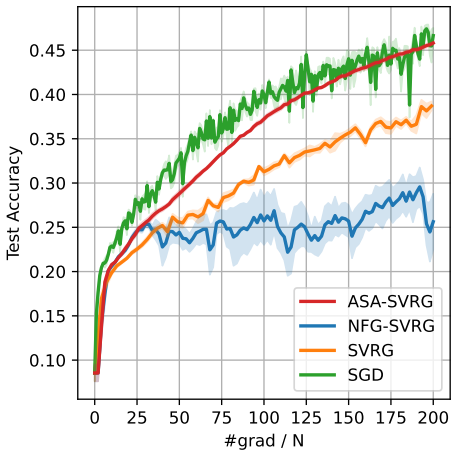
mini-batch size 512, whereas for mini-batch size 32 the training of ASA-SVRG collapsed as the inner-loop length K increased, and SGD itself also became unstable. Details are given in Appendix B.3.

5. Conclusions

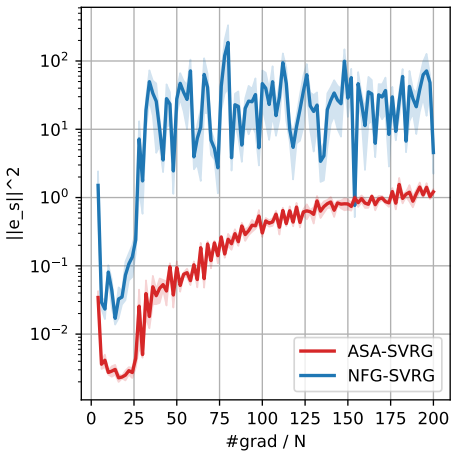
In this paper, we proposed Averaged Snapshots and Approximated full gradients for the stochastic variance-reduced gradient method (ASA-SVRG) as a variance reduction method for finite-sum minimization. The proposed method replaces the full-gradient computation by an average of stochastic gradients and uses the average of the parameters obtained in the inner loop as the snapshot, thereby reducing the approximation error of the full gradient. This design incorporates the insight of iterate averaging into the SVRG framework and theoretically minimizes the snapshot-dependent part of the upper bound on the error floor while retaining the computational efficiency of NFG-SVRG. Since the choice of the snapshot affects the deviation between the snapshot and the inner-loop trajectory, and hence the snapshot-dependent part of this upper bound, and the average parameter minimizes this deviation, for any inner-loop run, among snapshots formed from that run, ASA-SVRG is a natural choice of snapshot. Since ASA-SVRG returns the average parameter, noise reduction similar to that of Polyak–Ruppert averaging can be expected. In addition, because the average parameter is reset at every outer loop, the average late in training consists only of parameters near the optimum, which may lead to a better solution. The contributions of this paper are threefold. First, we proposed ASA-SVRG, which uses the average parameter as the snapshot. Second, we evaluated the convergence of the proposed and related methods under a unified workflow. Third, we showed that the approximation error of the full gradient is bounded by the sum of the variance of the stochastic gradients and the deviation between the snapshot and the inner-loop trajectory, and that the average parameter minimizes this deviation among the snapshots formed from the same inner-loop run. The theoretical analysis revealed that SVRG converges linearly, that NFG-SVRG and ASA-SVRG converge to an error floor caused by the approximation error, and that the average snapshot minimizes the snapshot-dependent part of the upper bound on the approximation error of ASA-SVRG. In the experiments, the approximation error of ASA-SVRG was observed to be smaller than that of NFG-SVRG on the Mushroom dataset (a strongly convex setting). Moreover, on CIFAR-10 and ImageWoof (nonconvex settings), the approximation error of NFG-SVRG grew so large that training broke down, whereas ASA-SVRG avoided the full-gradient computation and kept the approximation error at a



(A)



(B)



(C)

Fig. 3: Results on ImageWoof: (A) training loss, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

practically negligible level. In particular, on ImageWoof, ASA-SVRG reached accuracy comparable to that of SGD with a similar number of gradient evaluations and with stability across iterations, and its accuracy fluctuated less from epoch to epoch than that of SGD. The smaller approximation error observed for ASA-SVRG is accompanied by better training behavior even in settings that violate the assumptions of the theoretical analysis.

Future work includes extending the convergence guarantees to nonconvex objectives and establishing practical guidelines for setting the inner-loop length K and the learning rate η. It also includes theoretically analyzing term (i) itself (the variance of the stochastic gradients), as well as evaluating how the choice of the snapshot affects the stochastic-gradient variance of subsequent epochs. As shown in Appendix B.2, in a nonconvex setting, decreasing the mini-batch size and thereby increasing the inner-loop length K can make the approximation error explode and destabilize the training of variance reduction methods including ASA-SVRG, owing to the lack of the contraction that exists in strongly convex settings. On the other hand, we confirmed that the proposed method works well experimentally even in settings that violate the assumptions of the theoretical analysis, such as the image classification on CIFAR-10 and ImageWoof, which are nonconvex and use mini-batch learning. Providing a theoretical justification of this experimental finding is also left for future work.

Data availability

The code and experimental results are available at https://github.com/yryo1005/ASAI_SVRG_programs.git.

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Conflicts of interest

The authors declare no competing interests.

Author contribution

- Conceptualization: Ryo Yamatomi, Suguru Asai, Hiroshi Ninomiya.
- Methodology: Ryo Yamatomi, Suguru Asai, Hiroshi Ninomiya.
- Investigation: Ryo Yamatomi, Suguru Asai, Hiroshi Ninomiya.
- Supervision: Hiroshi Ninomiya.
- Writing – original draft: Ryo Yamatomi, Suguru Asai, Hiroshi Ninomiya.
- Writing – review & editing: Ryo Yamatomi, Suguru Asai, Hiroshi Ninomiya.

Appendix

A. Proofs of lemmas

Lemma 1 (Upper bound on the expected squared norm of gradient differences) Under Assumptions 1 and 2, for any $\mathbf{w}$,

$$\mathbb{E}_n\left[\|\nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{w}^*)\|^2\right] \le 2L\big(f(\mathbf{w}) - f(\mathbf{w}^*)\big) \tag{25}$$

holds.

Proof. We introduce the auxiliary function, which is minimized at $\mathbf{w} = \mathbf{w}^*$,

$$\phi_n(\mathbf{w}) := f_n(\mathbf{w}) - f_n(\mathbf{w}^*) - \langle \nabla f_n(\mathbf{w}^*), \mathbf{w} - \mathbf{w}^* \rangle \tag{A-1}$$

Then

$$\nabla \phi_n(\mathbf{w}) = \nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{w}^*) \tag{A-2}$$

where $\phi_n(\mathbf{w}^*) = 0$. Moreover, by Assumption 2, $\phi_n(\mathbf{w}) \ge 0$ holds. By Assumption 1, for any candidate learning rate $\eta > 0$,

$$\begin{aligned} 0 &= \phi_n(\mathbf{w}^*) \\ &\le \phi_n\big(\mathbf{w} - \eta \nabla \phi_n(\mathbf{w})\big) \\ &\le \phi_n(\mathbf{w}) - \eta \|\nabla \phi_n(\mathbf{w})\|^2 + \frac{L}{2}\eta^2 \|\nabla \phi_n(\mathbf{w})\|^2 \end{aligned} \tag{A-3}$$

holds. Consider the right-hand side as a function of η,

$$\psi(\eta) := \phi_n(\mathbf{w}) - \eta \|\nabla \phi_n(\mathbf{w})\|^2 + \frac{L}{2}\eta^2 \|\nabla \phi_n(\mathbf{w})\|^2 \tag{A-4}$$

and setting $\frac{\mathrm{d}\psi(\eta)}{\mathrm{d}\eta} = 0$ gives the optimal value $\eta^* = 1/L$. Substituting η^* into Eq. (A-4) yields

$$0 \le \psi(\eta^*) = \phi_n(\mathbf{w}) - \frac{1}{2L}\|\nabla \phi_n(\mathbf{w})\|^2. \tag{A-5}$$

Therefore

$$\begin{aligned} &\|\nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{w}^*)\|^2 \\ &\quad \le 2L\Big(f_n(\mathbf{w}) - f_n(\mathbf{w}^*) - \langle \nabla f_n(\mathbf{w}^*), \mathbf{w} - \mathbf{w}^* \rangle\Big) \end{aligned} \tag{A-6}$$

holds. Averaging over all data $n = 1, \dots, N$ gives

$$\begin{aligned}
&\frac{1}{N}\sum_{n=1}^{N}\|\nabla f_n(\mathbf{w}) - \nabla f_n(\mathbf{w}^*)\|^2 \\
&\quad\le 2L\frac{1}{N}\sum_{n=1}^{N}\Big(f_n(\mathbf{w}) - f_n(\mathbf{w}^*) - \langle\nabla f_n(\mathbf{w}^*), \mathbf{w}-\mathbf{w}^*\rangle\Big) \\
&\quad= 2L\big(f(\mathbf{w}) - f(\mathbf{w}^*)\big) \qquad \text{(A-7)}
\end{aligned}$$

Here we used $\frac{1}{N}\sum_{n=1}^{N}\nabla f_n(\mathbf{w}^*) = \nabla f(\mathbf{w}^*) = \mathbf{0}$. Hence Eq. (25) holds. □

Lemma 2 (Second moment of the corrected gradient) Under Assumptions 1 and 2,

$$\begin{aligned}
&\mathbb{E}_{n_s^k}\big[\|\mathbf{v}_s^k\|^2\big] \\
&\quad\le 6L\big(f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big) \\
&\qquad + 18L\big(f(\mathbf{z}_s) - f(\mathbf{w}^*)\big) + 6\|\mathbf{e}_s\|^2 \qquad (26)
\end{aligned}$$

holds.

Proof. In Eq. (22), define

$$\begin{aligned}
\mathbf{a} &= \nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{w}^*), \\
\mathbf{b} &= \nabla f_{n_s^k}(\mathbf{z}_s) - \nabla f_{n_s^k}(\mathbf{w}^*), \\
\mathbf{c} &= \mathbf{g}_s \qquad \text{(A-8)}
\end{aligned}$$

and then

$$\mathbf{v}_s^k = \mathbf{a} - \mathbf{b} + \mathbf{c} \qquad \text{(A-9)}$$

holds. By the inequality $\|\mathbf{a}-\mathbf{b}+\mathbf{c}\|^2 \le 3\|\mathbf{a}\|^2+3\|\mathbf{b}\|^2+3\|\mathbf{c}\|^2$,

$$\begin{aligned}
&\mathbb{E}_{n_s^k}\big[\|\mathbf{v}_s^k\|^2\big] \\
&\quad\le 3\mathbb{E}_{n_s^k}\big[\|\nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{w}^*)\|^2\big] \\
&\qquad + 3\mathbb{E}_{n_s^k}\big[\|\nabla f_{n_s^k}(\mathbf{z}_s) - \nabla f_{n_s^k}(\mathbf{w}^*)\|^2\big] \\
&\qquad + 3\|\mathbf{g}_s\|^2 \qquad \text{(A-10)}
\end{aligned}$$

holds. By Eq. (25), the first and second terms on the right-hand side are bounded as

$$\begin{aligned}
&3\mathbb{E}_{n_s^k}\big[\|\nabla f_{n_s^k}(\mathbf{w}_s^k) - \nabla f_{n_s^k}(\mathbf{w}^*)\|^2\big] \\
&\qquad\le 6L\big(f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big), \qquad \text{(A-11)} \\
&3\mathbb{E}_{n_s^k}\big[\|\nabla f_{n_s^k}(\mathbf{z}_s) - \nabla f_{n_s^k}(\mathbf{w}^*)\|^2\big] \\
&\qquad\le 6L\big(f(\mathbf{z}_s) - f(\mathbf{w}^*)\big) \qquad \text{(A-12)}
\end{aligned}$$

Furthermore, since $\mathbf{g}_s = \mathbf{e}_s + \nabla f(\mathbf{z}_s)$,

$$\begin{aligned}
\|\mathbf{g}_s\|^2 &= \|\mathbf{e}_s + \nabla f(\mathbf{z}_s)\|^2 \\
&\le 2\|\mathbf{e}_s\|^2 + 2\|\nabla f(\mathbf{z}_s)\|^2 \\
&= 2\|\mathbf{e}_s\|^2 + 2\|\nabla f(\mathbf{z}_s) - \nabla f(\mathbf{w}^*)\|^2 \\
&\le 2\|\mathbf{e}_s\|^2 + 4L\big(f(\mathbf{z}_s) - f(\mathbf{w}^*)\big) \qquad \text{(A-13)}
\end{aligned}$$

holds. Here, the last inequality follows from Eq. (19), a consequence of Assumptions 1 and 2. Therefore

$$3\|\mathbf{g}_s\|^2 \le 6\|\mathbf{e}_s\|^2 + 12L\big(f(\mathbf{z}_s) - f(\mathbf{w}^*)\big) \qquad \text{(A-14)}$$

holds. Substituting Eqs. (A-11), (A-12), and (A-14) into Eq. (A-10) yields Eq. (26). □

Lemma 3 (Upper bound on the cumulative expected objective gap) Under $0 < \eta < 1/(3L)$,

$$\begin{aligned}
&c_1\sum_{k=0}^{K-1}\mathbb{E}\big[f(\mathbf{w}_s^k) - f(\mathbf{w}^*)\big] \\
&\quad\le \left(Kc_2 + \frac{2}{\mu}\right)\mathbb{E}\big[f(\mathbf{z}_s) - f(\mathbf{w}^*)\big] \\
&\qquad + Kc_3\mathbb{E}\big[\|\mathbf{e}_s\|^2\big] \qquad (28)
\end{aligned}$$

holds.

Proof. By the update rule $\mathbf{w}_s^{k+1} = \mathbf{w}_s^k - \eta\mathbf{v}_s^k$,

$$\begin{aligned}
&\mathbb{E}_{n_s^k}\big[\|\mathbf{w}_s^{k+1} - \mathbf{w}^*\|^2\big] \\
&= \mathbb{E}_{n_s^k}\big[\|\mathbf{w}_s^k - \mathbf{w}^* - \eta\mathbf{v}_s^k\|^2\big] \\
&= \|\mathbf{w}_s^k - \mathbf{w}^*\|^2 - 2\eta\big\langle\mathbb{E}_{n_s^k}[\mathbf{v}_s^k], \mathbf{w}_s^k - \mathbf{w}^*\big\rangle \\
&\quad + \eta^2\mathbb{E}_{n_s^k}\big[\|\mathbf{v}_s^k\|^2\big] \\
&= \|\mathbf{w}_s^k - \mathbf{w}^*\|^2 - 2\eta\langle\nabla f(\mathbf{w}_s^k), \mathbf{w}_s^k - \mathbf{w}^*\rangle \\
&\quad - 2\eta\langle\mathbf{e}_s, \mathbf{w}_s^k - \mathbf{w}^*\rangle + \eta^2\mathbb{E}_{n_s^k}\big[\|\mathbf{v}_s^k\|^2\big] \qquad \text{(A-15)}
\end{aligned}$$

Here, Eq. (23) was used. By Assumption 3,

$$\langle\nabla f(\mathbf{w}_s^k), \mathbf{w}_s^k - \mathbf{w}^*\rangle \ge f(\mathbf{w}_s^k) - f(\mathbf{w}^*) + \frac{\mu}{2}\|\mathbf{w}_s^k - \mathbf{w}^*\|^2 \qquad \text{(A-16)}$$

holds. Also, by the Cauchy–Schwarz inequality and Young's inequality,

$$\begin{aligned}
-2\eta\langle\mathbf{e}_s, \mathbf{w}_s^k - \mathbf{w}^*\rangle &\le 2\eta\|\mathbf{e}_s\|\,\|\mathbf{w}_s^k - \mathbf{w}^*\| \\
&\le \eta\mu\|\mathbf{w}_s^k - \mathbf{w}^*\|^2 + \frac{\eta}{\mu}\|\mathbf{e}_s\|^2 \qquad \text{(A-17)}
\end{aligned}$$

holds. When we substitute Eqs. (A-16) and (A-17) into Eq. (A-15), the terms $\eta\mu\|\mathbf{w}_s^k - \mathbf{w}^*\|^2$ cancel, yielding

$$\begin{aligned}
&\mathbb{E}_{n_s^k}\left[\|\mathbf{w}_s^{k+1}-\mathbf{w}^*\|^2\right] \\
&\le \|\mathbf{w}_s^k-\mathbf{w}^*\|^2-2\eta\big(f(\mathbf{w}_s^k)-f(\mathbf{w}^*)\big) \\
&\quad+\frac{\eta}{\mu}\|\mathbf{e}_s\|^2+\eta^2\mathbb{E}_{n_s^k}\left[\|\mathbf{v}_s^k\|^2\right] \qquad \text{(A-18)}
\end{aligned}$$

Substituting Eq. (26) further, we obtain

$$\begin{aligned}
&\mathbb{E}_{n_s^k}\left[\|\mathbf{w}_s^{k+1}-\mathbf{w}^*\|^2\right] \\
&\le \|\mathbf{w}_s^k-\mathbf{w}^*\|^2-2\eta\big(f(\mathbf{w}_s^k)-f(\mathbf{w}^*)\big) \\
&\quad+6\eta^2L\big(f(\mathbf{w}_s^k)-f(\mathbf{w}^*)\big) \\
&\quad+18\eta^2L\big(f(\mathbf{z}_s)-f(\mathbf{w}^*)\big) \\
&\quad+\left(6\eta^2+\frac{\eta}{\mu}\right)\|\mathbf{e}_s\|^2 \\
&= \|\mathbf{w}_s^k-\mathbf{w}^*\|^2-c_1\big(f(\mathbf{w}_s^k)-f(\mathbf{w}^*)\big) \\
&\quad+c_2\big(f(\mathbf{z}_s)-f(\mathbf{w}^*)\big)+c_3\|\mathbf{e}_s\|^2 \qquad \text{(A-19)}
\end{aligned}$$

When $0<\eta<1/(3L)$, $c_1>0$ holds. Taking the total expectation of both sides of Eq. (A-19) and using the tower property gives

$$\begin{aligned}
&\mathbb{E}\left[\|\mathbf{w}_s^{k+1}-\mathbf{w}^*\|^2\right] \\
&\le \mathbb{E}\left[\|\mathbf{w}_s^k-\mathbf{w}^*\|^2\right]-c_1\mathbb{E}\left[f(\mathbf{w}_s^k)-f(\mathbf{w}^*)\right] \\
&\quad+c_2\mathbb{E}\left[f(\mathbf{z}_s)-f(\mathbf{w}^*)\right]+c_3\mathbb{E}\left[\|\mathbf{e}_s\|^2\right] \qquad \text{(A-20)}
\end{aligned}$$

Summing Eq. (A-20) over $k=0,1,\ldots,K-1$ and telescoping the distance terms yields,

$$\begin{aligned}
&c_1\sum_{k=0}^{K-1}\mathbb{E}\left[f(\mathbf{w}_s^k)-f(\mathbf{w}^*)\right] \\
&\le \mathbb{E}\left[\|\mathbf{w}_s^0-\mathbf{w}^*\|^2\right]-\mathbb{E}\left[\|\mathbf{w}_s^K-\mathbf{w}^*\|^2\right] \\
&\quad+Kc_2\mathbb{E}\left[f(\mathbf{z}_s)-f(\mathbf{w}^*)\right]+Kc_3\mathbb{E}\left[\|\mathbf{e}_s\|^2\right] \qquad \text{(A-21)}
\end{aligned}$$

Using $-\mathbb{E}[\|\mathbf{w}_s^K-\mathbf{w}^*\|^2]\le 0$, $\mathbf{w}_s^0=\mathbf{z}_s$, and Eq. (21), we obtain

$$\mathbb{E}\left[\|\mathbf{w}_s^0-\mathbf{w}^*\|^2\right]\le\frac{2}{\mu}\mathbb{E}\left[f(\mathbf{z}_s)-f(\mathbf{w}^*)\right]. \qquad \text{(A-22)}$$

Combining this bound with Eq. (A-21) and dropping the nonpositive term $-\mathbb{E}[\|\mathbf{w}_s^K-\mathbf{w}^*\|^2]$ yields Eq. (28). □

B. Additional experiments

This appendix presents additional experiments: a9a classification (B.1), and CIFAR-10 and ImageWoof classification with different mini-batch sizes (B.2 and B.3).

B.1 Binary classification on the a9a dataset

This subsection presents the results of two binary classification experiments on the a9a dataset. We first present an experiment that reproduces the supplementary experiment (Appendix A.1, least squares regression) of the NFG-SVRG paper [6], the prior work, using the mean squared error (MSE) as the loss function. Unlike the experiment in Section 4.1, this experiment considers a small-scale problem that does not satisfy the assumptions of the theoretical analysis. The purpose of this experiment is not to correspond to the theoretical analysis, but to confirm that the implementations of SVRG, NFG-SVRG, and the proposed ASA-SVRG operate correctly and consistently with the results of the prior work. We then present a second experiment that matches the loss function and the model to those of the Mushroom experiment in Section 4.1, using the binary cross-entropy (BCE) loss.

We use the a9a dataset published in LIBSVM Data. The number of samples is $N=32561$ and the feature dimension is $D=123$ (each feature is a binary variable taking 0 or 1) [17]. All data are split 9:1 into training and validation sets, with $N_{\text{train}}=29304$ training samples and $N_{\text{test}}=3257$ validation samples. No preprocessing such as standardization is applied to the features. The target labels are the original labels -1 and $+1$ converted to 0 and 1, respectively.

Setting with the MSE loss Following Appendix A.1 (Eq. (8)) of the NFG-SVRG paper, the model uses the least-squares error of a linear combination passed through the sigmoid function. The objective function is defined as

$$f(\mathbf{w})=\frac{1}{N}\sum_{n=1}^{N}\big(y_n-h_n(\mathbf{w})\big)^2,\quad h_n(\mathbf{w})=\sigma(\mathbf{w}^\top\mathbf{x}_n) \qquad \text{(B-1)}$$

where σ is the sigmoid function, $\mathbf{x}_n\in\{0,1\}^D$ is the feature vector, and $y_n\in\{0,1\}$ is the target label. As in the NFG-SVRG paper, no regularization term is used in this experiment. Because of the composition with the sigmoid function, Eq. (B-1) is nonconvex and does not satisfy Assumptions 2 and 3. The smoothness constant L is computed as $L=\kappa_{\max}\max_n\|\mathbf{x}_n\|^2$, where the upper bound $\kappa_{\max}$ on the eigenvalues of the Hessian of a single-sample loss is obtained by numerical search over $z\in[-30,30]$. Computing this numerically from the a9a data gives $L=2.1568$.

We compare four methods: SGD, SVRG, NFG-SVRG, and ASA-SVRG. The inner-loop length is $K=N_{\text{train}}=29304$, and the training data are passed over

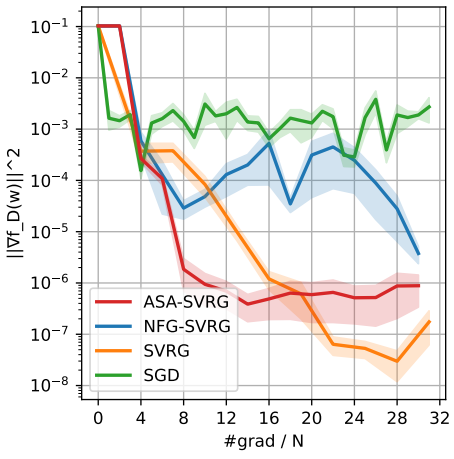
once per epoch by random reshuffling (online learning with mini-batch size one). As in Section 4.1, the learning rate is $\eta = 0.99/(3L) = 1.530030 \times 10^{-1}$, close to the upper bound of the condition $0 < \eta < 1/(3L)$ of Lemma 3. This value is determined analytically from the data and L, and no experimental tuning is performed. The number of outer loops S of each method was set so that #grad$/N$ is approximately 30. As in the experiments of the main text, the horizontal axis is the number of gradient computations #grad$/N$.

Figure B-1(A) shows the squared norm $\|\nabla f(\mathbf{z}_s)\|^2$ of the true full gradient at the end of each outer loop. SGD converges to an error floor under a constant learning rate and stalls at a certain level. In contrast, SVRG, NFG-SVRG, and ASA-SVRG all reduce $\|\nabla f(\mathbf{z}_s)\|^2$ substantially. This trend is broadly consistent with that reported in the NFG-SVRG paper [6].

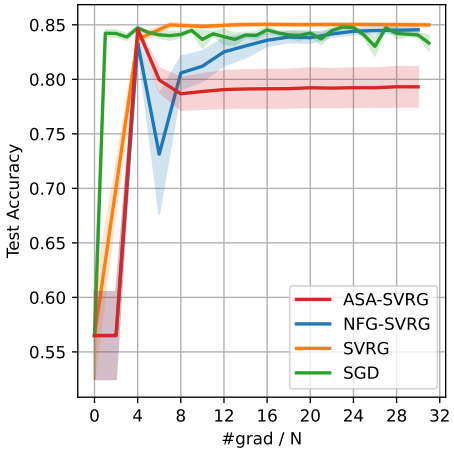
Figure B-1(B) shows the classification accuracy on the validation data. At the final epoch, the accuracies of the four methods reach similar levels, while NFG-SVRG and ASA-SVRG show somewhat larger variability across trials than SVRG.

Figure B-1(C) shows the approximation error $\|\mathbf{e}_s\|^2$ of ASA-SVRG and NFG-SVRG. In the first half of training, the approximation error of ASA-SVRG is consistently lower than that of NFG-SVRG, with a difference of several orders of magnitude. As training proceeds, both methods decrease to the level of rounding error and approach the approximation error of SVRG (identically 0). This behavior is qualitatively consistent with the effect of reducing the deviation between the snapshot and the inner-loop trajectory (Theorem 3); the theoretical results on the deviation (Theorem 2 and Theorem 3) apply since each f_n is smooth, whereas Theorem 1 does not, because the problem is nonconvex.

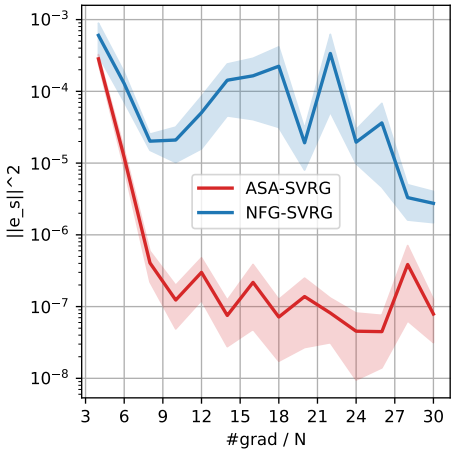
Setting with the BCE loss In this setting, we match the loss function and the model to those of the Mushroom experiment in Section 4.1 and change only the dataset to a9a. That is, the model is a linear combination model with an intercept b, and binary classification is performed by logistic regression. The objective function is the binary cross-entropy loss with L2 regularization on both the weight $\mathbf{w}$ and the intercept b, defined in exactly the same form as Eq. (48) in Section 4.1. Regularizing the intercept b as well makes μ-strong convexity with $\mu = \lambda$ hold uniformly for all parameters $(\mathbf{w}, b)$, so that Assumption 3 is strictly satisfied. As in Section 4.1, the smoothness constant is computed as $L = \frac{1}{4} \max_n \|\tilde{\mathbf{x}}_n\|^2 + \lambda$, where $\tilde{\mathbf{x}}_n$ is the feature vector augmented with a constant 1



(A)



(B)



(C)

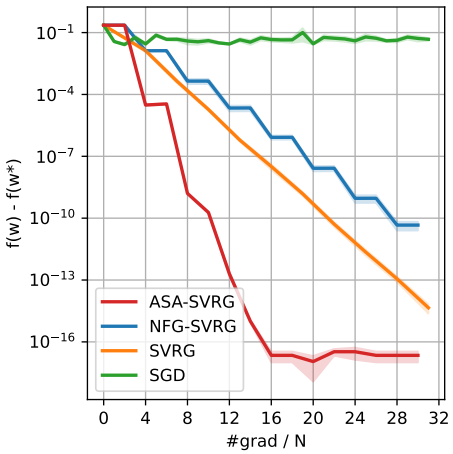
Fig. B-1: Results on the a9a dataset with the MSE loss: (A) squared norm of the full gradient $\|\nabla f(\mathbf{z}_s)\|^2$, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

corresponding to the intercept, and the regularization coefficient λ is set using the same ratio, 3×10^{-2},

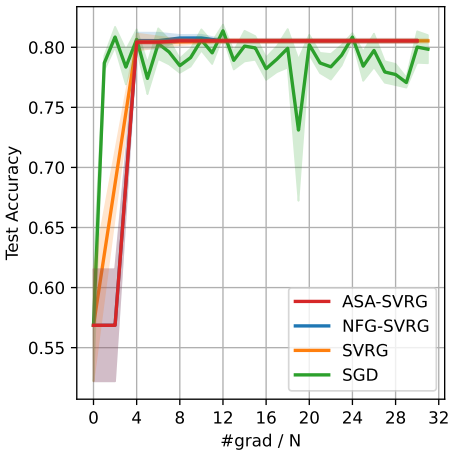
as in Section 4.1. Each record of a9a is a 123-dimensional feature vector ($D = 123$) in which exactly 14 entries equal 1 and the remaining entries equal 0, so $\max_n \|\tilde{\mathbf{x}}_n\|^2 = 15$, which gives $L = 3.8625$ and $\lambda = \mu = 0.1125$ (the condition number $\mu/L \approx 0.029$ matches that of Section 4.1).

We compare four methods: SGD, SVRG, NFG-SVRG, and ASA-SVRG. The inner-loop length is $K = N_{\text{train}} = 29304$, and, as in Section 4.1, the training data are passed over once per epoch by random reshuffling (online learning with mini-batch size one). The learning rate is $\eta = 0.99/(3L) = 8.543689 \times 10^{-2}$, which is close to the upper bound of the condition $0 < \eta < 1/(3L)$ of Lemma 3. The optimal value $f(\mathbf{w}^*)$ is estimated with high accuracy by L-BFGS-B. The number of outer loops S of each method was set so that #grad$/N$ is approximately 30, the same target value as in Section 4.1.

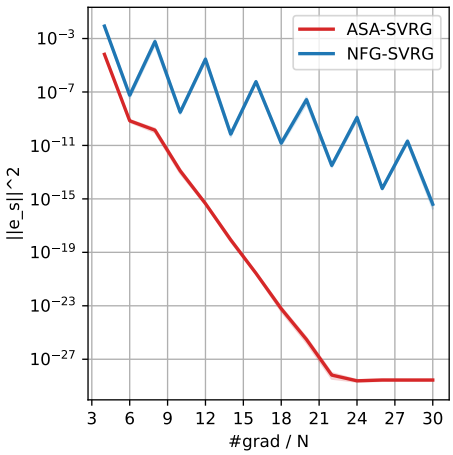
Figure B-2(A) shows the objective error $f(\mathbf{z}_s) - f(\mathbf{w}^*)$. Figure B-2(B) shows the classification accuracy on the validation data. Figure B-2(C) shows the approximation error $\|\mathbf{e}_s\|^2$ of the full gradient for ASA-SVRG and NFG-SVRG. Although the dataset changes from the Mushroom dataset, with 7311 samples and 22 features, to a9a, with 29304 samples and 123 features, the trends in (A)–(C) are nearly the same as those in Section 4.1, which is consistent with the effect of reducing the deviation between the snapshot and the inner-loop trajectory (Theorem 3): the approximation error of SVRG remains identically zero (Lemma 4), and that of ASA-SVRG is observed to be smaller than that of NFG-SVRG, on another dataset that satisfies the assumptions of the theoretical analysis.



(A)



(B)



(C)

Fig. B-2: Results on the a9a dataset with the BCE loss: (A) objective error $f(\mathbf{z}_s) - f(\mathbf{w}^*)$, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

B.2 Classification on CIFAR-10 with different mini-batch sizes

Section 4.2 reports the results for mini-batch size 128 on the CIFAR-10 classification experiment with AlexNet; this subsection presents the results of additional experiments in which the mini-batch size is instead changed to 32 and 512. The problem setting, model, learning rate, and so on are the same as in Section 4.2, and only the mini-batch size is changed. Since the inner-loop length K is inversely proportional to the mini-batch size, $K = 1563$ for mini-batch size 32 and $K = 98$ for mini-batch size 512 (compared with $K = 391$ for mini-batch size 128 in Section 4.2). The number of outer loops S of each method was set so that #grad$/N$ is approximately 20 for mini-batch size 32 and approximately 350 for mini-batch size 512.

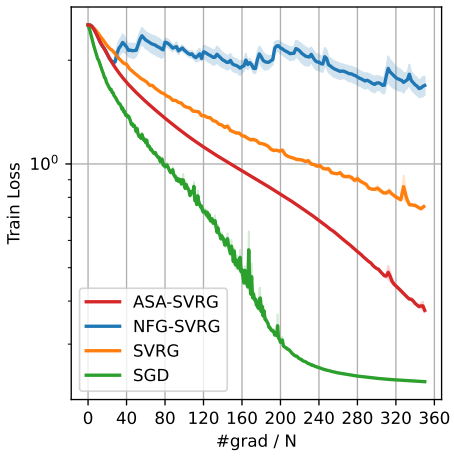
Figure B-3(A)–(C) show the training loss, the classification accuracy on the validation data, and the approximation error $\|\mathbf{e}_s\|^2$ for mini-batch size 512. At

the final epoch, as in Section 4.2 (mini-batch size 128), SGD, SVRG, and ASA-SVRG reach almost the same accuracy. In contrast, the accuracy of NFG-SVRG is far below these, and its approximation error is also large.

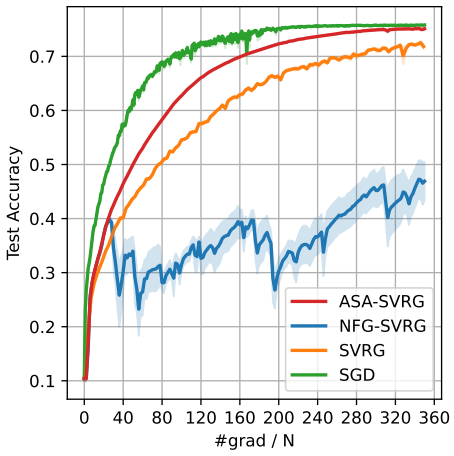
Figure B-4(A)–(C) show the training loss, the classification accuracy on the validation data, and the approximation error for mini-batch size 32.

SGD and SVRG improve their classification accuracy stably as training proceeds. In contrast, ASA-SVRG improves its accuracy up to a point, but from $\#\mathrm{grad}/N \approx 8$ onward its training loss diverges rapidly and its validation accuracy drops to the level of random guessing for 10-class classification, so that training collapses. For ASA-SVRG, training was stopped when this divergence was observed. The validation accuracy of NFG-SVRG also gradually drops to the level of random guessing, but its training loss stays almost constant and does not diverge as that of ASA-SVRG does, indicating that its parameters are hardly updated and its training is essentially stagnant. Figure B-4(C) shows that in the first half of training ($\#\mathrm{grad}/N \lesssim 6$) the approximation error of ASA-SVRG is lower than that of NFG-SVRG, but after $\#\mathrm{grad}/N \approx 8$, where ASA-SVRG diverges, the approximation error of NFG-SVRG is far lower than that of ASA-SVRG. However, as described above, the small approximation error of NFG-SVRG in this phase results from the training of NFG-SVRG itself being almost stagnant, with the parameters hardly changing from the snapshot, and does not mean that NFG-SVRG provides a good approximation.

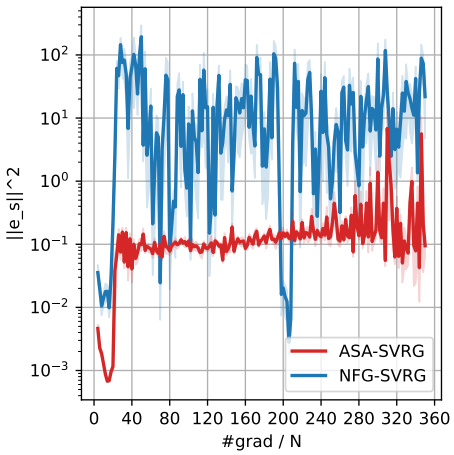
We state below a hypothesis on why decreasing the mini-batch size increases the approximation error and destabilizes training. As shown in the proof of Theorem 2 in Section 3, the snapshot-dependent term (ii) of the upper bound on the approximation error is the right-hand side of Eq. (44), $\frac{2L^2}{K}\sum_{k=0}^{K-1}\mathbb{E}[\|\mathbf{w}_{s-1}^k - \mathbf{z}_s\|^2]$, which is governed by the inner-loop length K. Since the parameters $\mathbf{w}_{s-1}^k$ obtained at each inner-loop step gradually deviate from the snapshot $\mathbf{z}_s$, the average magnitude of the individual terms $\|\mathbf{w}_{s-1}^k - \mathbf{z}_s\|^2$ can also increase as K grows. Therefore, even with the normalization by $1/K$, this upper bound does not necessarily decrease monotonically with K. In the first experiment (Section 4.1), online learning with mini-batch size one is used and thus $K = N_{\mathrm{train}} = 7311$ is large, but as noted in Section 4.1, the experimental step size gives $\rho > 1$, so the contraction bound in Theorem 1 does not directly explain this observed stability. In contrast, the nonconvex setting targeted by the present experiment (CIFAR-10 classification with AlexNet) has no such guarantee of contraction; hence our hypothesis is that decreasing



(A)

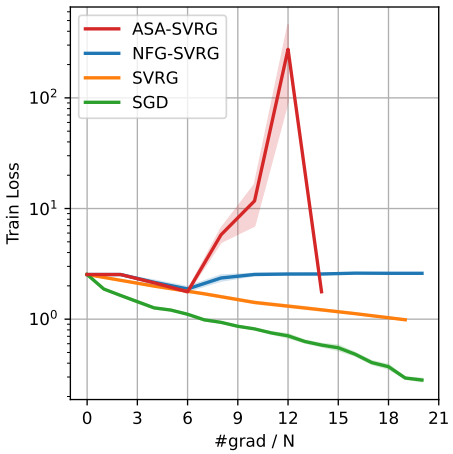


(B)

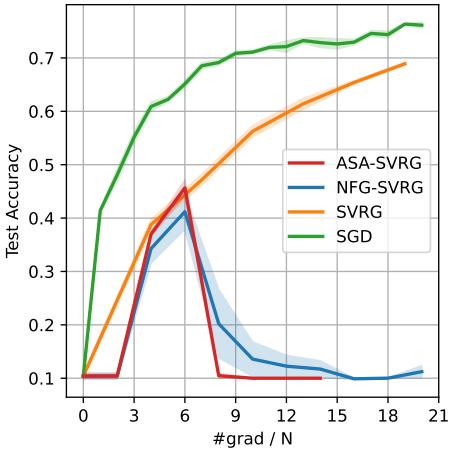


(C)

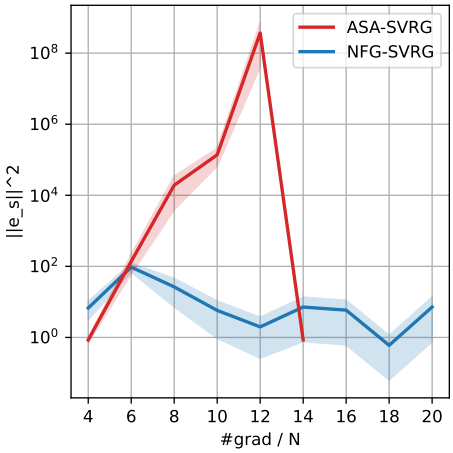
Fig. B-3: Results for mini-batch size 512: (A) training loss, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$



(A)



(B)



(C)

Fig. B-4: Results for mini-batch size 32: (A) training loss, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

the mini-batch size and increasing K leaves this upper bound unsuppressed and tends to lead to an explosion of the approximation error and unstable training. A theoretical verification of this hypothesis is left for future work.

B.3 Classification on ImageWoof with different mini-batch sizes

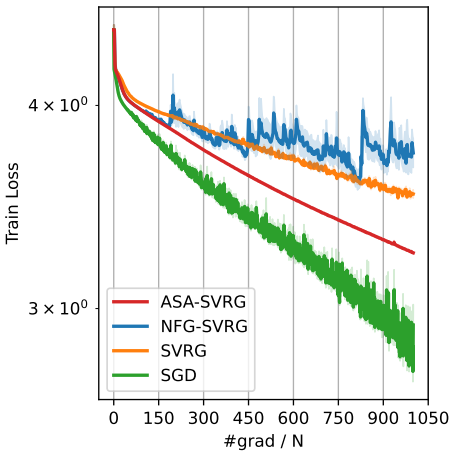
Section 4.3 reports the results for mini-batch size 128 on the ImageWoof classification experiment with ResNet18; this subsection presents the results of additional experiments in which the mini-batch size is instead changed to 512 and 32. The problem setting, model, learning rate, and so on are the same as in Section 4.3, and only the mini-batch size is changed. Since the inner-loop length K is inversely proportional to the mini-batch size, $K = 18$ for mini-batch size 512 and $K = 283$ for mini-batch size 32 (compared with $K = 71$ for mini-batch size 128 in Section 4.3). The number of outer loops S of each method was set so that $\#\text{grad}/N$ is approximately 1000 for mini-batch size 512 and approximately 30 for mini-batch size 32.

Figure B-5(A)–(C) show the training loss, the classification accuracy on the validation data, and the approximation error $\|\mathbf{e}_s\|^2$ for mini-batch size 512.

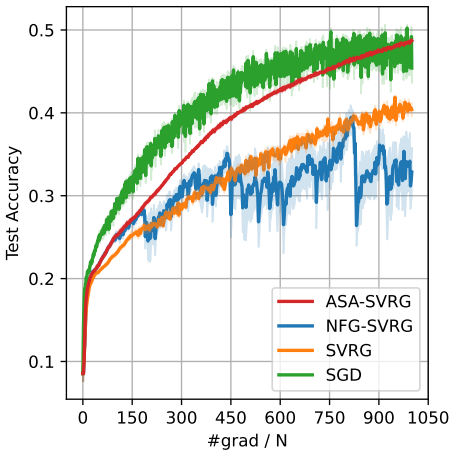
Relative to mini-batch size 128 in Section 4.3 and to mini-batch size 32 discussed below, the accuracy of NFG-SVRG here is comparatively high and improves at a level close to that of SVRG until the middle of training. In the latter half of training, however, it falls below SVRG and shows large variability across trials. The final accuracy of ASA-SVRG reaches the same level as that of SGD and exceeds those of SVRG and NFG-SVRG. Moreover, the epoch-to-epoch fluctuation of the accuracy of ASA-SVRG is markedly smaller than that of the other methods, including SGD, and its accuracy improves smoothly. Except in the early stage of training, the approximation error of ASA-SVRG is far below that of NFG-SVRG.

Figure B-6(A)–(C) show the training loss, the classification accuracy on the validation data, and the approximation error for mini-batch size 32.

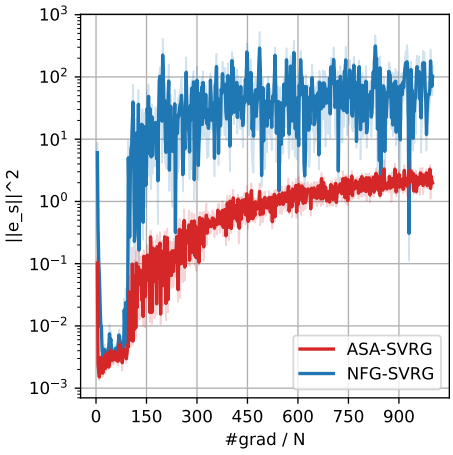
For mini-batch size 32, the training of SGD, which had improved the accuracy fastest in the previous experiments, becomes unstable: both the training loss and the accuracy fluctuate widely, and no stable improvement is observed. ASA-SVRG improves its accuracy up to the middle of training at least as fast as that of SVRG and far faster than that of SGD, but around $\#\text{grad}/N = 30$ its training loss starts to increase and its accuracy drops, so that training collapses. The accuracy of NFG-SVRG improves in the early stage of training but decreases from around $\#\text{grad}/N = 10$ onward, falling below the accuracies of SVRG and ASA-



(A)

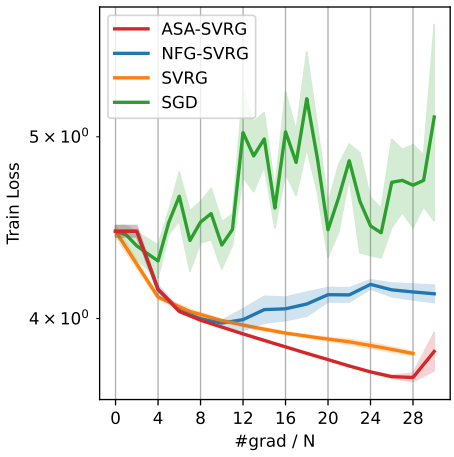


(B)

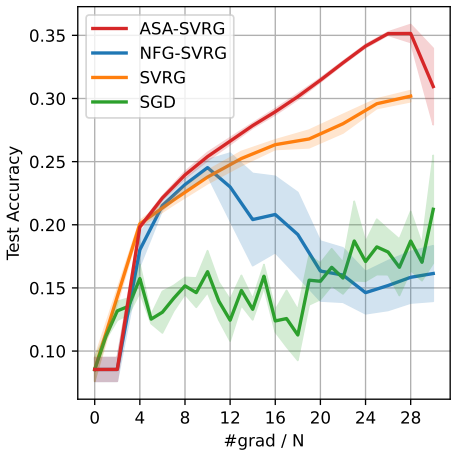


(C)

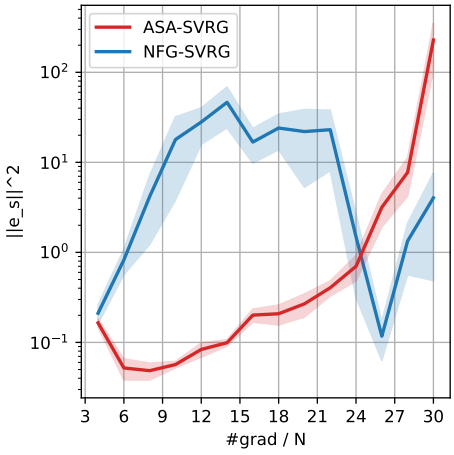
Fig. B-5: Results on ImageWoof for mini-batch size 512: (A) training loss, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$



(A)



(B)



(C)

Fig. B-6: Results on ImageWoof for mini-batch size 32: (A) training loss, (B) classification accuracy on the validation data, and (C) approximation error $\|\mathbf{e}_s\|^2$

SVRG. Figure B-6(C) shows that the approximation error of ASA-SVRG is far below that of NFG-SVRG in the first half of training, but increases sharply from around #grad$/N$ = 25 onward, which corresponds to the collapse of training. This is consistent with the tendency in the CIFAR-10 experiment with mini-batch

size 32 in Appendix B.2, and with the hypothesis that decreasing the mini-batch size and thereby increasing the inner-loop length K destabilizes the training of ASA-SVRG through the growth of the approximation error.

On the other hand, across every dataset and mini-batch size examined in Section 4 and in Appendices B.2 and B.3, the training of SVRG never breaks down and always proceeds stably. This is presumably because SVRG uses the full gradient and hence has no approximation error.

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